

*Review Article*STRUCTURE–ANTIOXIDANT ACTIVITY RELATIONSHIPS OF
FLAVONOIDS AND PHENOLIC ACIDS

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Abstract—The recent explosion of interest in the bioactivity of the the flavonoids of higher plants is due, at least in part, to the potential health benefits of these polyphenolic components of major dietary constituents. This review article discusses the biological properties of the flavonoids and focuses on the relationship between their antioxidant activity, as hydrogen donating free radical scavengers, and their chemical structures. This culminates in a proposed hierarchy of antioxidant activity in the aqueous phase. The cumulative findings concerning structure–antioxidant activity relationships in the lipophilic phase derive from studies on fatty acids, liposomes, and low-density lipoproteins; the factors underlying the influence of the different classes of polyphenols in enhancing their resistance to oxidation are discussed and support the contention that the partition coefficients of the flavonoids as well as their rates of reaction with the relevant radicals define the antioxidant activities in the lipophilic phase.

Keywords—Flavonoid, Antioxidant, Total antioxidant activity, Catechin, Low-density lipoprotein, Anthocyanidin, ABTS, Tea, Wine

INTRODUCTION

The polyphenolic flavonoids have the diphenylpropane ($C_6C_3C_6$) skeleton. The family includes monomeric flavanols, flavanones, anthocyanidins, flavones, and flavonols. Along with the phenylpropanoids or hydroxycinnamic acid derivatives (C_6C_3), flavonols and to a lesser extent flavones are found in almost every plant.^{1–3} While flavanones and flavones are often found together (e.g., in citrus fruits) and are connected by specific enzymes, there is a certain mutual exclusion between flavones and flavonols in many plant families and anthocyanins are almost absent in flavanone-rich plants.⁴ The interconnections between the individual flavonoid subgroups are shown in Fig. 1. Individual differences within each group result from the variation in number and arrangement of the hydroxyl groups as well as from the nature and extent of alkylation and/or glycosylation of these groups. The most commonly occurring flavones and flavonols are those with dihydroxylation in the 3' and 4' positions of the B ring, and

to a lesser extent, those with a lone B ring-hydroxyl group in the 4' position. The preferred glycosylation site on the flavonoids is the 3 position and less frequently the 7 position. Glucose is the most usual sugar residue but others include galactose, rhamnose, xylose. The most common hydroxycinnamic acids are not present in plants in a free state but occur most frequently as simple esters with quinic acid or glucose.⁵ They are not glycosylated at the phenolic hydroxyl groups.

It is well-known that diets rich in fruit and vegetables are protective against cardiovascular disease and certain forms of cancer,^{6,7} and perhaps against other diseases also. These protective effects have been attributed, in large part, to the antioxidants present including the antioxidant nutrients vitamin C and β -carotene, but also the minor carotenoids, and plant phenolics such as the flavonoids and phenylpropanoids may also have a significant role. The polyphenolic components of higher plants may act as antioxidants or as agents of other mechanisms contributing to anticarcinogenic or cardioprotective action. The flavonoids constitute a large class of compounds, ubiquitous in plants, containing a number of phenolic hydroxyl groups attached

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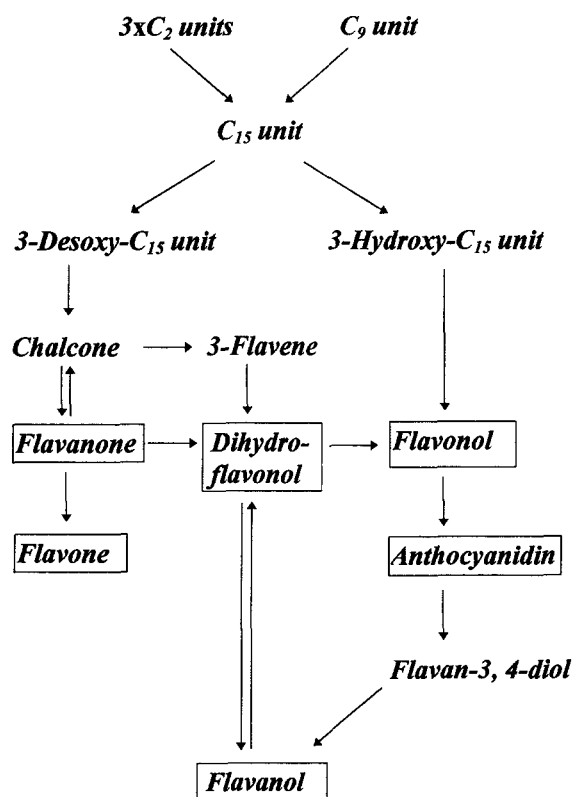


Fig. 1. Interconnections of flavonoid subgroups.

to ring structures, conferring the antioxidant activity.⁸ They often occur in the glycosidic form, cleavage of the free polyphenol taking place possibly in the gastrointestinal tract. Plant polyphenols are multifunctional and can act as reducing agents, hydrogen donating antioxidants, and singlet oxygen quenchers. In some cases metal chelation properties have been proposed. Estimates of daily intake range from about 20 mg to 1 g.⁹ The flavanols, particularly the catechin and catechin-gallate ester family, and the flavonols quercetin, kaempferol, and their glycosides are constituents of the beverages green and black teas and red wine (Table 1). Quercetin is also a predominant component of onions and apples, and myricetin and quercetin of berries. The flavanones are mainly found in citrus fruits.

In support of flavonoids exerting a protective effect *in vivo* are the findings of a Dutch epidemiological study showing that coronary heart disease in elderly males is inversely correlated with their intake of flavonoids.⁹ Most of their dietary flavonoids derived from tea (48% of the flavonoid intake), onions (29%), apples (7%), and red wine (1%). The risk of death from coronary heart disease in the lower tertile of flavonoid intake was about 2.4 times that of the upper tertile. It still remains to be established to what extents the antioxidant and antithrombotic properties of the polyphenols contribute to this protection.

The constituents of red wine are factors of particular interest due to the intrigue created by the French paradox. The Southern French have a very low incidence of coronary heart disease despite their high fat diet and smoking tendencies.¹⁰ One of the features that has been highlighted relates to the high consumption of red wine by the French and the question as to whether the polyphenolic antioxidants from this dietary source contribute to protection from coronary heart disease along with the antioxidants in olive oil and the high intake of antioxidant nutrients from the fresh fruit and vegetable-rich Mediterranean diet.

The chemical properties of polyphenols in terms of the availability of the phenolic hydrogens as hydrogen-donating radical scavengers predicts their antioxidant activity. For a polyphenol to be defined as an antioxidant it must satisfy two basic conditions: first, when present in low concentration relative to the substrate to be oxidized it can delay, retard, or prevent the autoxidation or free radical-mediated oxidation;¹¹ second, the resulting radical formed after scavenging must be stable—through intramolecular hydrogen bonding on further oxidation.¹²

The biological, pharmacological, and medicinal properties of the flavonoids have been extensively reviewed.^{13–15} Flavonoids and other plant phenolics are reported, in addition to their free radical scavenging activity,¹⁶ to have multiple biological activities^{17,18} including vasodilatory,^{19,20} anticarcinogenic, antiinflammatory, antibacterial, immune-stimulating, antiallergic, antiviral, and estrogenic effects, as well as being inhibitors of phospholipase A₂, cyclooxygenase, and lipoxygenase^{17,21–27} (about which a considerable amount of work has been published), glutathione reductase,²⁸ and xanthine oxidase.²⁹ Earlier reports of the carcinogenic activity³⁰ of quercetin in bracken and ferns have not been substantiated. The biological activities of the phenylpropanoids³¹ and their role as antimicrobial agents^{32,33} are well-recognized in addition to their properties as antiallergic and antiinflammatory agents through lipoxygenase inhibition³⁴ and their antimutagenic actions.^{35,36} The polyphenols have also been reported to elicit antiviral activities against HIV,^{37–39} Herpes simplex,⁴⁰ influenza virus,⁴¹ and Rhinovirus.⁴² Polyphenols can act as inhibitors of cyclin-dependent kinases from breast carcinoma cells.⁴³ Quercetin has also been shown to mediate the downregulation of mutant p53 in a human breast cancer cell line⁴⁴ and other studies indicate that quercetin-induced growth-inhibitory activity in ovarian cancer cells may be mediated by modulation of transforming growth factor beta 1 production.⁴⁵

The chemistry of the flavonoids is predictive of their free radical scavenging activity because the reduction potentials of flavonoid radicals are lower than those of alkyl peroxy radicals and the superoxide radical,

Table 1. Some Dietary Sources of Flavonoids

Flavanol	
Epicatechin	
Catechin	
Epigallocatechin	green and black teas
Epicatechin gallate	red wine
Epigallocatechin gallate	
Flavanone	
Naringin	peel of citrus fruits
Taxifolin	citrus fruits
Flavonol	
Kaempferol	endive, leek, broccoli, radish, grapefruit, black tea
Quercetin	onion, lettuce, broccoli, cranberry, apple skin, berries, olive, tea, red wine
Myricetin	cranberry, grapes, red wine
Flavone	
Chrysin	fruit skin
Apigenin	celery, parsley
Anthocyanidins	
Malvidin	red grapes, red wine
Cyanidin	cherry, raspberry, strawberry, grapes
Apigenidin	coloured fruit and peels
Phenyl propanoids	
Ferulic acid	wheat, corn, rice, tomatoes, spinach, cabbage, asparagus
Caffeic acid	white grapes, white wine, olives, olive oil, spinach, cabbage, asparagus, coffee
<i>p</i> -Coumaric acid	white grapes, white wine, tomatoes, spinach, cabbage, asparagus
Chlorogenic acid	apples, pears, cherries, plums, peaches, apricots, blueberries, tomatoes, anis

which means the flavonoids may inactivate these oxyl species and prevent the deleterious consequences of their reactions.^{24,46} Their antioxidant activity is also reported as scavengers of superoxide radical,^{47–50} although there is conflicting evidence,^{51,52} peroxy radical scavengers,^{53,54} inhibitory effects on lipid peroxidation;^{55–57} inhibition of LDL oxidation induced by copper ions and macrophages^{58,59} with half-maximal inhibition induced by compounds ranging in concentration from 1–10 μ M. In studies on model systems, the catechins from tea have revealed high activity in erythrocyte membranes and in rat liver microsomes with greatest protection from lipid oxidation by epigallocatechin gallate and epicatechin gallate, the latter being 10 times more effective than vitamin E.⁶⁰ Others have evaluated the antioxidant activity of flavonoids by investigating their effects on cells in culture: pretreatment of cells followed by exposure to reactive oxygen species resulted in proposed concentrations required for protection being in the order of flavanols greater than flavonols. Administration of flavonoids to mice prior to whole body gamma irradiation showed a good relationship between anticlastogenic activity in vivo and antioxidant effects in vitro.⁶¹

Although there is a wealth of data on the importance

of antioxidants in conferring stability towards or protection from oxidation, the correlation between antioxidant activity and chemical structure is far from clear. Different methods of assessment, varying substrate systems, and differential concentrations of active antioxidants all have contributed to the confounding of the issue. A rapid screening assay has been developed to determine the antioxidant potencies of natural and synthetic antioxidants.⁶² It is based on the interaction between linoleic acid and an azo compound as the initiator of peroxidation and the modulation of conjugated diene hydroperoxide absorption at 234 nm. [The azo-initiator used is 2,2'-azobis (2-amidino propane) dihydrochloride (ABAP), although 2,2'-azobis (2,4-dimethyl valerionitrile) (AMVN) is commonly used in lipophilic systems.] The theory underlying the test is based on the assumption that the lipophilic inhibitors scavenge lipid peroxy radicals at the surface of the micelle. The application of this method results in the broad categorization of antioxidants into three classes: a) compounds that are better antioxidants than α -tocopherol; b) compounds (generally analogs of vitamins E and C) that are as effective as α -tocopherol; and c) some compounds that are less effective as antioxidants than α -tocopherol, examples being quercetin, probucol, and dihydroxybenzoic acid in this lipophilic system.

While this method has its uses for studying peroxy radical scavengers, it is not sufficiently sensitive for defining detailed structure–activity relationships nor for assessing aqueous phase interactions.

Many in vitro studies have defined the antioxidant potential of the polyphenols as direct radical scavengers and as agents capable of enhancing the resistance to oxidation of low density lipoproteins implicated in the pathogenesis of coronary heart disease. All the major polyphenolic constituents of food, flavonols such as quercetin and kempferol, flavones such as luteolin, flavanols including catechins, and anthocyanidins, for example, cyanidin and malvidin and their glycosides, show greater efficacy in these systems as antioxidants on a mole for mole basis than the antioxidant nutrients vitamin C, vitamin E, and β -carotene,^{63–65} which are readily absorbed in the main.

SPECTROSCOPIC IDENTIFICATION AND STRUCTURAL CHARACTERISTICS OF PHENOLICS

It is well known^{66,67} that most flavones and flavonols exhibit two major absorption bands in the ultraviolet/visible region, Band I in the 320–385 nm range representing the B ring absorption, and Band II in the 250–285 nm range representing A ring absorption. The data are consistent with the recorded observations that increase in the numbers of hydroxyl groups induce a red shift (Table 2), for instance, from 367 nm in kaemp-

ferol (with hydroxyl groups in positions 3,5,7,4') to 371 nm in quercetin (3,5,7,3',4') to 374 nm in myricetin (3,5,7,3',4',5'). The absence of a 3-OH group in flavones (which distinguishes flavones from flavonols) means that Band I is always at a shorter wavelength by 20–30 nm than that in the equivalent flavonols, for example, apigenin (5,7,4') 337 nm, kaempferol (3,5,7,4') 367 nm. O-Methylation and glycosylation produce hypsochromic shifts.

Flavanones have a saturated heterocyclic C ring; the consequent lack of conjugation between the A and B rings, in contrast to the flavones and flavonols, is defined by their UV spectral characteristics as well as in their lowered antioxidant activity. The UV spectra of flavanones (e.g., naringenin 5,7,4') or dihydroflavonols (e.g., taxifolin 3,5,7,3',4') are very characteristic in that they exhibit a very strong maximum (Band II) between 270 and 295 nm, namely, 288 and 285 nm, respectively, and only a shoulder for Band I at 326, 327 nm, respectively. Band II appears as one peak (ca 270 nm) in compounds with a monosubstituted B ring, but as two peaks or one peak (ca 258 nm) plus a shoulder (at about 272 nm) when a di-, tri-, or O-substituted B ring is present.⁶⁶ The color of the anthocyanins varies according to the number and position of the hydroxyl groups⁶⁸ and they show distinctive Band I peaks in the 450–560 nm region (due to the B ring hydroxy cinnamoyl system) (refs) and Band II in the 240–280 nm region due to the A ring benzoyl system.⁶⁶

ANTIOXIDANT POTENTIALS OF POLYPHENOLS AGAINST RADICALS GENERATED IN THE AQUEOUS PHASE AND STRUCTURE–ACTIVITY RELATIONSHIPS

The assay for the total antioxidant activity (TAA), or the Trolox equivalent antioxidant activity (TEAC), measures the concentration of Trolox solution with an equivalent antioxidant potential to a standard concentration of the compound under investigation.^{69,70} The TEAC reflects the ability of hydrogen-donating antioxidants to scavenge the ABTS^{•+} radical cation, absorbing in the near-ir region at 734, 645, and 815 nm (Fig. 2), compared with that of Trolox, the water-soluble vitamin E analog. Antioxidants suppress the absorbance at 734 nm to an extent and on a time scale dependent on the antioxidant activity. The TEAC is defined as the concentration of Trolox solution with equivalent antioxidant potential to a 1 mM concentration of the compound under investigation.^{69,70}

Comparison of 3,5,7,3',4'-pentahydroxy polyphenolic structures

The structures in Fig. 3 represent three flavonoids: quercetin, a pentahydroxy flavanol (flavone-3-ol); cat-

Table 2. Relationship Between Hydroxyl Group Arrangements and Absorption Maxima

	OH Arrangement	Band Positions
Flavonols		
Kaempferol	3, 5, 7, 4'	367 nm
Quercetin	3, 5, 7, 3', 4'	371 nm
Myricetin	3, 5, 7, 3', 4', 5'	374 nm
Flavones		
Chrysin	5, 7	313 nm
Apigenin	5, 7, 4'	337 nm
Flavanones		
Naringenin	5, 7, 4'	289 (326 sh) nm
Taxifolin	3, 5, 7, 3', 4'	290 (327 sh) nm

echin, a flavanol (3-hydroxyflavan); and cyanidin, an anthocyanidin, with identical arrangements of the five hydroxyl groups and their Trolox equivalent antioxidant activities. Quercetin has an identical number of hydroxyl groups in the same positions as catechin but also contains the 2,3-double bond in the C ring and the 4-oxo function. This structure advantage confers an enhancement of the TEAC value to 4.7 ± 0.1 mM compared to the saturated heterocyclic ring of catechin with approximately half the antioxidant activity (2.4 ± 0.05 mM)⁶³ (Table 3). Cyanidin with the central anthocyanidin C ring, allowing conjugation, has approximately the same antioxidant activity as quercetin (4.4 ± 0.12 mM). The results demonstrate the importance of the unsaturation in the C ring and allowing electron delocalization across the molecule for stabilization of the aryloxyl radical.

Role of 3-OH group on the C ring and its relationship with the unsaturation in the ring

The glycosylation of flavonoids reduces their activity when compared to the corresponding aglycones¹² (Table 3). Blocking the 3-hydroxyl group in the C ring of quercetin as a glycoside (while retaining the 3',4'-dihydroxy structure in the B ring) as in rutin, or quercetin rutinoside (Fig. 4a), or removing the 3-OH group in the C ring as in luteolin decreases the antioxidant activity to a value of 2.4 ± 0.12 mM ($n = 7$) and 2.1 ± 0.05 ($n = 4$), respectively. Thus, the maximum effectiveness for radical scavenging apparently requires the 3-OH group attached to the 2,3-double bond and adjacent to the 4-carbonyl in the C ring. Retaining the catechol-type structure in the B ring but removing the 2-3 double bond in the C ring, eliminates the means of delocalization of electrons from the aryloxyl radical on the B ring to the A ring, as in taxifolin (dihydroquercetin), giving a TEAC value of 1.9 ± 0.03 mM ($n = 6$), of the same order as catechin but more effective than kaempferol. The comparison of quercetin with lute-

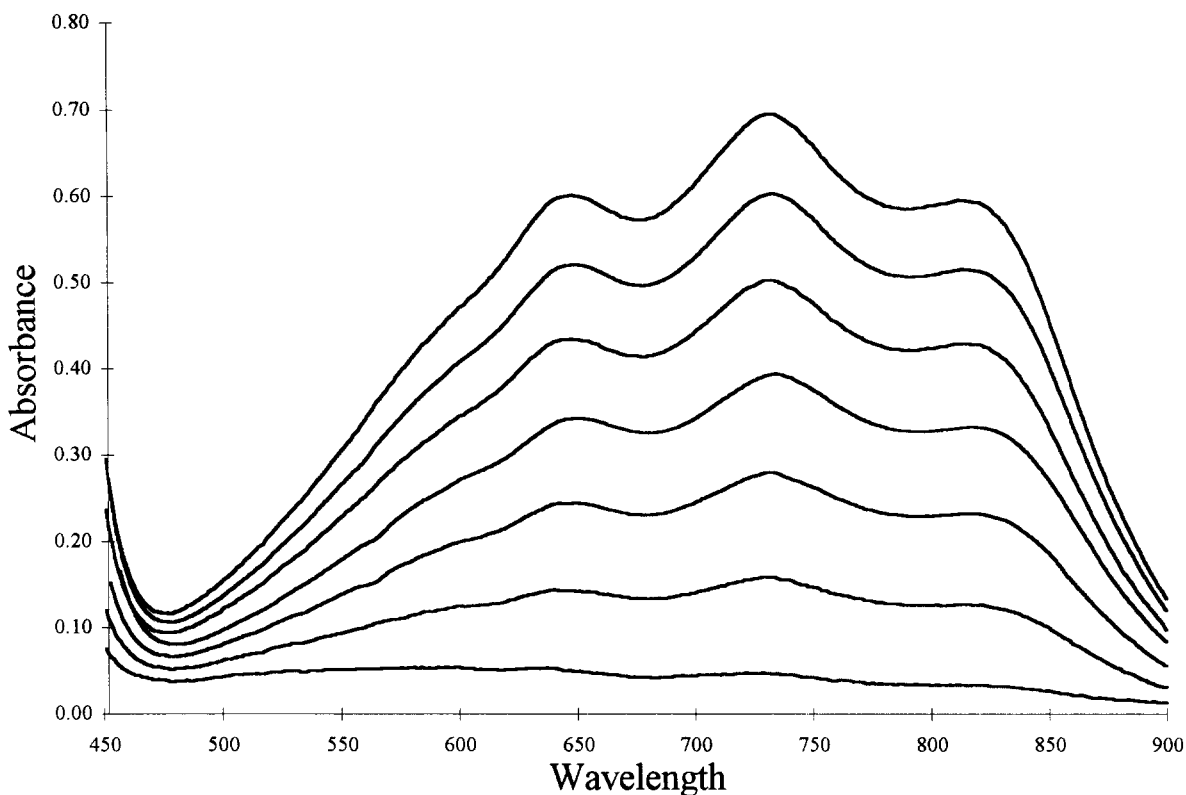


Fig. 2. Near infrared spectrum of the $\text{ABTS}^{\bullet+}$ radical cation, formed on interaction of ABTS (2,2'-azinobis-[3-ethyl benzothiazoline-6-sulphonic acid]) (final concentration $150 \mu\text{M}$) with activated metmyoglobin (final concentration $2.5 \mu\text{M}$, final concentration of hydrogen peroxide $75 \mu\text{M}$).

olin and rutin demonstrates the influence of the 3-OH in combination with the adjacent double bond in the C ring, and if one is dispensed with the other apparently loses its impact on the antioxidant activity. Furthermore, the reduction of the 2,3-unsaturated bond in the C ring of kaempferol, converting it to dihydrokaempferol, has no influence on the total antioxidant activity (1.39 ± 0.02 , $n = 3$) as compared with kaempferol substantiating the notion of the lack of effect of the specific structural characteristics in the C ring on the TAA in the absence of the orthodiphenolic structure in the B ring. Indeed, the single OH group in the B ring apparently makes little contribution and even in conjunction with the conjugated double bond system and the 3-OH group contributes, little to the antioxidant potential of kaempferol. This is supported by comparison with apigenin with no 3-OH or chrysin (Fig. 4b) with the absence of the 3-OH and the single 4'-OH on the B ring giving a similar TEAC of 1.4 mM . Indeed, the 2,3 double bond apparently contributes little to the hydrogen-donating ability without the diphenolic structure in the B ring because naringenin (dihydroapigenin) with a saturated heterocyclic ring and no 3-OH group has much the same TEAC value. Comparing catechin with the flavones, it is clear that the 4-keto group is only functional in conjugation with the 2-3 double bond.

Importance of the orthodiphenolic structure in the B ring

The importance of the adjacency of the two hydroxyl groups in the ortho-diphenolic arrangement in the B ring of quercetin to its antioxidant activity of 4.7 is revealed from a study of morin in which the dihydroxy groups are arranged meta to each other in the B ring, decreasing the value to 2.55, approximating to that of catechin (Fig. 5 and Table 3).

However, a related structure but with a lone 4'-OH group in the B ring, kaempferol, differing from quercetin in the absence of the 3'-OH group from the B ring, has just 27% of the latter's antioxidant activity ($1.34 \pm 0.08 \text{ mM}$, $n = 6$). Thus, presumably, the 2,3 double bond is not so relevant when the B ring lacks the *o*-dihydroxy arrangement and only contains one hydroxy substituent, because the monophenolic ring is not such an effective hydrogen donor. This value demonstrates the strong influence of the orthodiphenolic structure on the TAA.

The presence of a third OH group in the B ring does not enhance the effectiveness against aqueous phase radicals as in myricetin compared with quercetin (Fig. 5) and the anthocyanidin delphinidin compared with cyanidin (see Anthocyanin and Anthocyanidins

Table 3. Hierarchy of Trolox Equivalent Antioxidant Activities of Polyphenols

Compound	Free OH-Substituents	Glycosylated Position	TEAC (mM)	<i>n</i>	Family
Epicatechin gallate	3, 5, 7, 3', 4', 3'', 4'', 5''		4.9 ± 0.02	[3]	flavanol
Epigallocatechin gallate	3, 5, 7, 3', 4', 5', 3'', 4'', 5''		4.8 ± 0.06	[3]	flavanol
Quercetin	3, 5, 7, 3', 4'		4.7 ± 0.1	[6]	flavonol
Delphinidin	3, 5, 7, 3', 4', 5'		4.44 ± 0.11	[5]	anthocyanidin
Cyanidin	3, 5, 7, 3', 4'		4.4 ± 0.12	[5]	anthocyanidin
Epigallocatechin	3, 5, 7, 3', 4', 5'		3.8 ± 0.06	[3]	flavanol
Keracyanin	5, 7, 3', 4'	3-rut	3.25 ± 0.1	[3]	anthocyanin
Myricetin	3, 5, 7, 3', 4', 5'		3.1 ± 0.30	[6]	flavonol
Gallic acid	3, 4, 5		3.01 ± 0.05	[7]	hydroxybenzoate
Ideain	5, 7, 3', 4'	3-gal	2.9 ± 0.03	[3]	anthocyanin
Morin	3, 5, 7, 3', 4', 5'		2.55 ± 0.02	[3]	flavonol
Epicatechin	3, 5, 7, 3', 4'		2.5 ± 0.02	[6]	flavanol
Gallic acid methyl ester	3, 4, 5		2.44 ± 0.03	[3]	hydroxybenzoate
Catechin	3, 5, 7, 3', 4'		2.4 ± 0.05	[9]	flavanol
Rutin	5, 7, 3', 4', 5'	3-rut	2.4 ± 0.06	[7]	flavonol
Apigenidin	5, 7, 4'		2.35 ± 0.2	[4]	anthocyanidin
Peonidin	3, 5, 7, 4'	3'-OMe	2.22 ± 0.2	[4]	anthocyanidin
Luteolin	5, 7, 3', 4'		2.1 ± 0.05	[4]	flavone
Malvidin	3, 5, 7, 4'	3',5'-di-OMe	2.06 ± 0.1	[4]	anthocyanidin
Taxifolin	3, 5, 7, 3', 4'		1.9 ± 0.03	[6]	flavanone
Oenin	5, 7, 4'	3',5'-diOMe	1.78 ± 0.02	[3]	anthocyanin
Luteolin-4'-glucoside	5, 7, 3'	3-gluc 4'-gluc	1.74 ± 0.09	[4]	flavone
Naringenin	5, 7, 4'		1.53 ± 0.05	[4]	flavanone
Apigenin	5, 7, 4'		1.45 ± 0.08	[6]	flavone
Chrysin	5, 7		1.43 ± 0.07	[6]	flavone
Hesperitin	3, 5, 7, 3'	4'-OMe	1.37 ± 0.08	[3]	flavanone
Kaempferol	3, 5, 7, 4'		1.34 ± 0.08	[6]	flavonol
Pelargonidin	3, 5, 7, 4'		1.30 ± 0.1	[6]	anthocyanidin
Hesperidin	3, 5, 3'	4'-OMe 7-rut	1.08 ± 0.04	[5]	flavanone
Luteolin-3',7'-diglucoside	5, 4'	3',7'-digluc	0.79 ± 0.04	[4]	flavone
Narirutin	5, 4'	5-rut	0.76 ± 0.05	[3]	flavanone

section). This is also supported by the findings of Pokorny,⁷¹ who reported that the presence of three hydroxyl groups on the aromatic nucleus did not improve antioxidant efficiency. Robinetin and myricetin, with additional hydroxyl groups at the 5' position, have been suggested to have resulting enhanced antioxidant activities in lipid systems over those of their corresponding flavones that do not possess the 5'-hydroxyl group.⁷² As shown in oxidizing low-density lipoproteins,⁷³ the increase in the number of hydroxyl groups in the B ring of compounds with a saturated heterocyclic ring, for instance, epicatechin vs. epigallocatechin (see Antioxidant Activity of the Catechins and Catechin-Gallate Esters section) did not enhance the antioxidant activity in these lipophilic systems, but the converse was the case against aqueous phase radicals (an increase of 2.5 ± 0.02 to 3.8 ± 0.06).

The influence of B ring hydroxylation on the antioxidant activity of the flavones and flavanols

The manipulation of the —OH substitution in the B ring in flavones (with the 2,3-double bond with the

4-oxo function in the C ring, but no 3-OH group) allows the calculation of the contribution of the 5,7-dihydroxyphenolic arrangement in the A ring to the antioxidant activity. The contribution of the 3',4'-dihydroxy structure contributes about 25% to the antioxidant activity of luteolin with a TEAC value of 2.1 ± 0.05 mM, because dehydroxylation at the 3'-position as in apigenin (Fig. 6) decreases the value to 1.45 ± 0.08 , which is the same as the result for chrysin with unsubstituted B and C rings (1.43 ± 0.07) (Table 3). Thus, this value can reasonably be attributed to the antioxidant activity of the 5,7-metadihydroxy arrangement of the A ring. The flavanone, naringenin (dihydroapigenin or dihydro-3-desoxy kaempferol), with the same hydroxyl arrangement in the A ring but with only a single 4'-OH group in the B ring also has a TEAC value (1.5 ± 0.05) consistent with the anticipated contribution from the A ring. The findings for hesperetin (1.37 ± 0.08), with an identical structure to naringenin except for the 3-OH, 4-methoxy substitution in the B ring also confirm the attribution to the 5,7-dihydroxyphenolic A ring of a TEAC value in the range 1.35–1.5 mM. Flavanones with only one hydroxyl group in the B ring (naringenin and hesperetin)

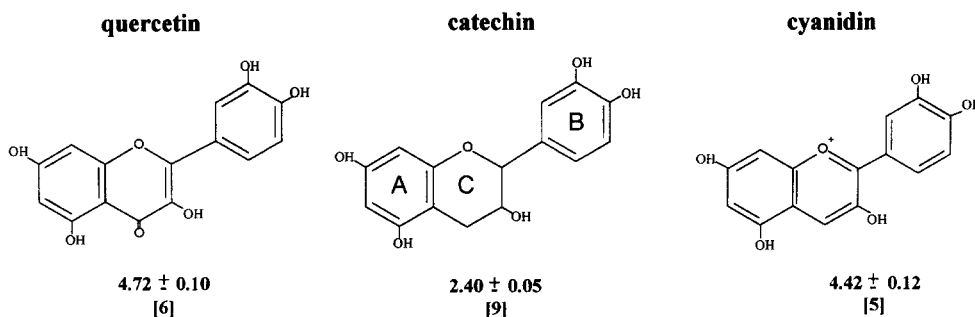


Fig. 3. Structure-antioxidant activity comparisons of 3,5,7,3',4'-pentahydroxy polyphenolic structures.

have been suggested to possess little antioxidant activity in lipid systems, and this is substantiated by further studies using stripped corn oil; the time to reach a peroxide value of 50 was greatest for the compounds with extra OH groups on the B ring:¹² robinetin > myricetin > quercetin = quercitrin = taxifolin > rhamnetin > rutin = naringenin > hesperetin = hesperidin.

However, interactions with aqueous phase radicals (Fig. 6) show that there are contributions to the antioxidant activity from hydroxyl groups on the A ring in the absence of the dihydroxy structure in the B ring.

Comparison of naringenin (TEAC 1.5 ± 0.05 mM) with narirutin (TEAC 0.76 ± 0.05) (Fig. 7) shows that glycosylation of the 7-OH group in a structure with a saturated heterocyclic ring and with a lone hydroxyl group on the B ring has a strongly suppressive influence on the antioxidant activity. Similar effects are observed when hesperetin (with a *p*-OH group in the B ring replaced by a methoxy and the 3'-OH group, in contrast to naringenin) (TEAC 1.4 ± 0.08 mM) is compared with its rhamnoside, hesperidin, which has a glycosylated 7-OH group (TEAC 1.08 ± 0.03 mM); the same effect is evident when luteolin (TEAC 2.09 ± 0.05) is compared with its 4'-mono and 3',7-diglucosides (TEAC 1.74 ± 0.09 and 0.79 ± 0.04).

Antioxidant activity of the catechins and catechin-gallate esters

The study of the catechins is important for understanding the antioxidant properties of teas. The structures of the catechins and catechin/gallate esters are shown in Fig. 8 with their relative antioxidant potentials against radicals in the aqueous phase, expressed as the Trolox equivalent antioxidant activity.⁷³ Catechins (including epicatechins) with three hydroxyl groups in the B ring are the gallocatechins and those esterified to gallic acid at the 3-OH group in the C ring are the catechin gallates. There is no electron delocalization between the A and B rings due to the saturation of

the heterocyclic ring; the antioxidant activity responds broadly to the tenet that the structures with the most hydroxyl groups exert the greatest antioxidant activity, with the catechin isomers at 2.4 and 2.5, more than twice as effective as vitamins E and C (TEAC = 1). The catechin-gallate esters reflect the contribution from gallic acid (3,4,5-trihydroxybenzoic acid). Quercetin (Fig. 3) has an identical number of hydroxyl groups in the same positions as catechin, but also contains the 2,3-double bond in the C ring and the 4-oxo group. This structural advantage confers an enhancement of the TEAC value to 4.7 ± 0.10 mM ($n = 6$). Thus, the catechin structure with a TEAC value of 2.4 ± 0.02 mM ($n = 6$) can be modified to enhance its antioxidant potential to 4.7 as in quercetin by incorporation of the 2,3-double bond and the 4-oxo function, both in the C ring and as in epigallocatechin gallate (4.75 ± 0.06 mM, $n = 9$) by ester linkage via the 3-OH group to gallic acid and incorporation of an additional 5'-OH group in the B ring (Table 3). The insertion of a third adjacent hydroxyl group in the B ring as in epigallocatechin enhances the antioxidant activity to 3.8 ± 0.06 mM ($n = 3$).

The antioxidant potentials of the tea catechins, on a molar basis, against radicals generated in the aqueous phase are, in order of decreasing effectiveness, epicatechin gallate $\approx$ epigallocatechin gallate > epigallocatechin > gallic acid > epicatechin $\approx$ catechin.

Quercetin, the flavonol with the same OH group arrangement as catechin, gave approximately twice this value due to the altered bonding in the C ring, allowing delocalization between the A and B rings stabilizing the aryloxy radical after hydrogen donation.

The contribution to the composition of the green tea extract (Table 4, line 1) of the catechin-gallate components is 26.7% dry weight of solids (compared with the flavonols and their glycosides at 6% of the total polyphenolic composition of green tea and ca. 15% unidentified polyphenols) composed of epigallocatechin gallate, 11.16%; epicatechin gallate, 2.25%; epigallocatechin, 10.32%; epicatechin, 2.45%; catechin, 0.53%.⁷⁴ The anti-

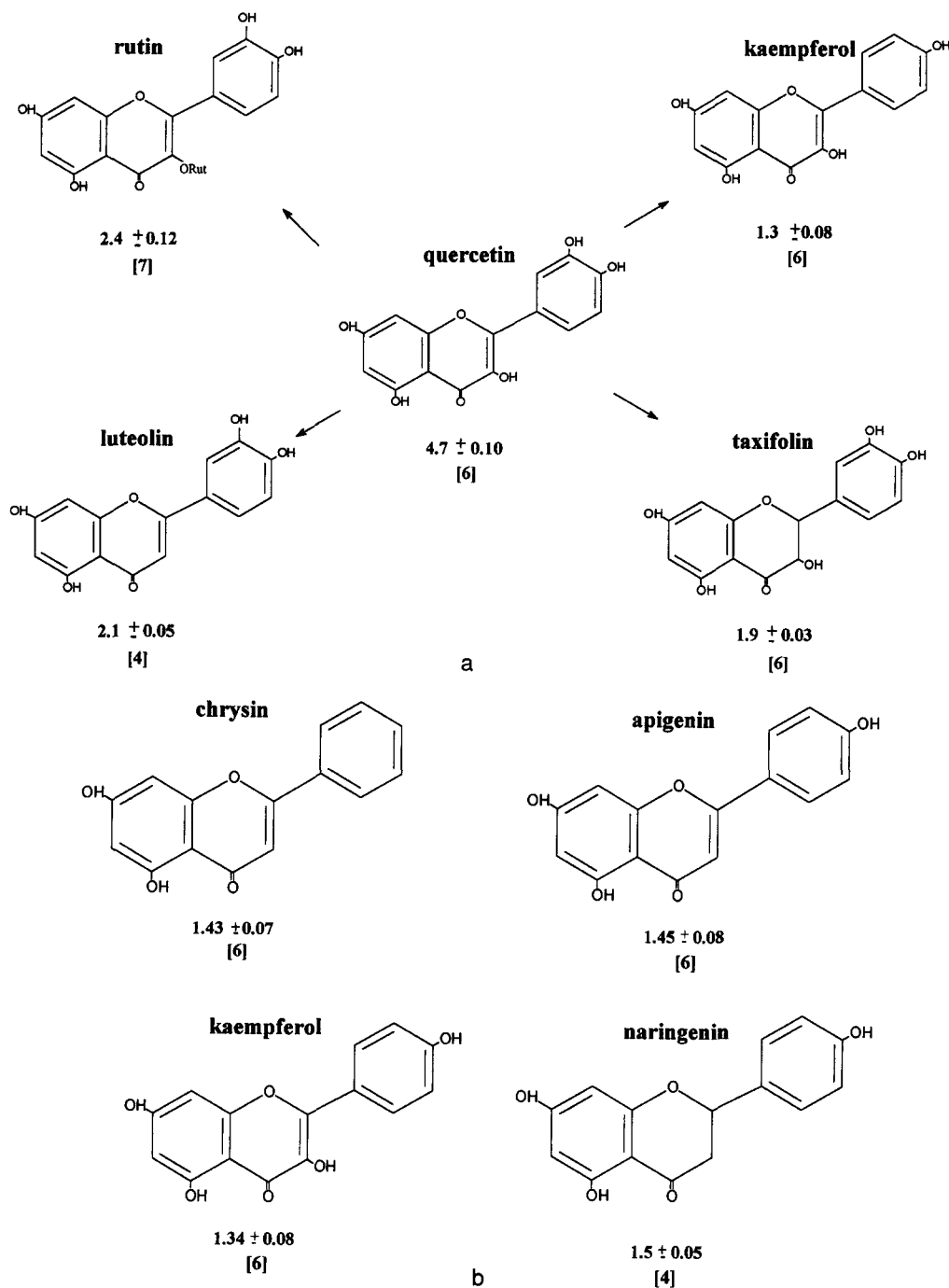


Fig. 4.(a, b) Influence of the 3-OH group with the unsaturated C₂—C₃ link on the antioxidant activity of flavonols.

oxidant capacities of tea polyphenolic constituents (Table 4, line 2) in relation to their concentrations in tea are used to calculate their predicted contributions to the antioxidant potential of green tea. The result of 2.95 mM is reasonably consistent with the measured TEAC of the proportionately combined catechin-gallate constituents with a value of 2.76 ± 0.06 mM (Table 5). Taking into account the antioxidant activity of the polyphenolic constituents of green tea in relation to their relative com-

positions, the order of contribution to the antioxidant effectiveness in green tea is epigallocatechin $\approx$ epigallocatechin gallate $\gg$ epicatechin gallate = epicatechin > catechin.

The green tea preparation itself (at 1000 ppm concentration) gave a TEAC value of 3.78 ± 0.03 mM ($n = 9$). Thus, 78% of the antioxidant activity of green tea extracts can be accounted for by the catechins and catechin-gallate esters from the calculated data and

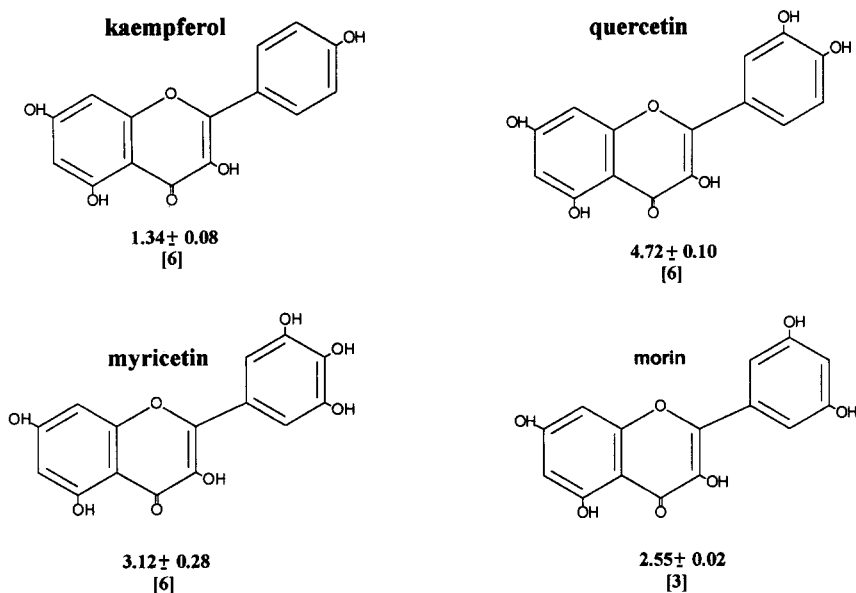


Fig. 5. The importance of the orthodiphenolic structure in the B ring to the antioxidant activity of the flavonols.

73% from the reconstitution experiment, the rest being the contribution from the unidentified polyphenols.⁷³

However, the total green tea polyphenol extract (44% of the dry weight of the green tea preparation) shows a total antioxidant activity of 3.36 (Table 5), which accounts for 90% of the antioxidant activity of the tea preparation. It is interesting to note that the anticarcinogenic actions of green tea and its constituents in vivo, particularly epigallocatechin gallate, have also been reported in human⁷⁵ and animal studies.^{76–78}

Although the total polyphenol content of black tea extract (44.94% by weight) is similar to that of green tea extract, only 6.9% by weight is comprised of catechin-gallate components. The rest includes theaflavins, thearubigens, and undefined polymeric polyphenols formed during fermentation. Interestingly, the antioxidant activity of black tea extract (3.49 ± 0.05 , $n = 12$) is very close to that of green tea (3.78 ± 0.03 mM, $n = 9$).⁷³ Indeed, theaflavin has been reported to inhibit Cu^{2+} -mediated LDL oxidation.⁷⁹

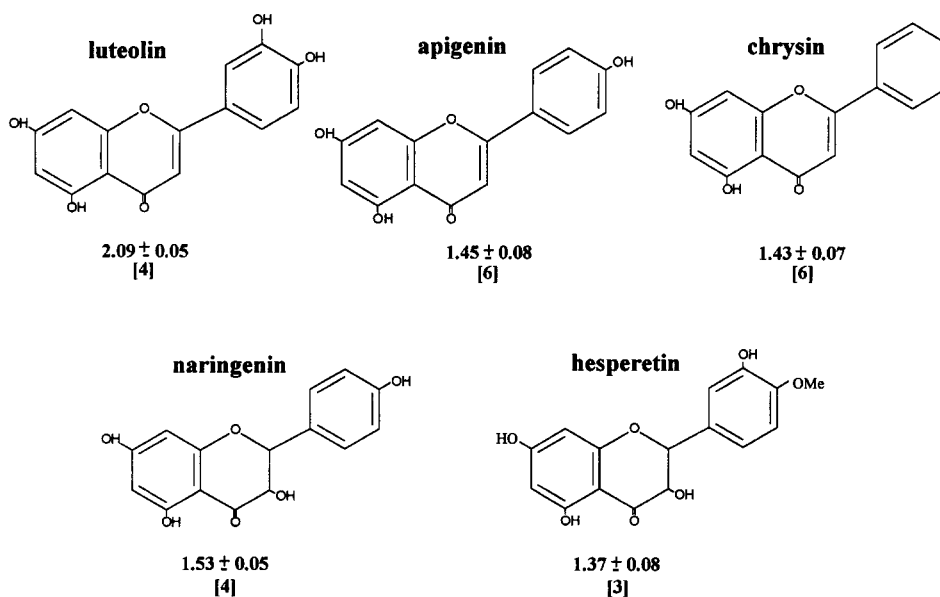


Fig. 6. The influence of the hydroxylation in the B ring on the antioxidant activity of the flavones and flavanols.

Anthocyanin and anthocyanidins

The major antioxidant activity of the anthocyanins (Fig. 9), pigments of berries and grape skins, can be ascribed again to the reducing power of the *o*-dihydroxy structure in the B ring as in cyanidin with a similar TEAC (4.4 ± 0.01 , $n = 5$) as quercetin, and the same number and arrangement of the five hydroxyl groups. Dehydroxylation to a monophenol in the B ring as in pelargonidin (1.3 ± 0.1 , $n = 6$) gives much the same value as the equivalent flavon-3-ol, kaempferol. Insertion of a methoxy group in the 3' position of the B ring with the 4'-hydroxy group in peonidin enhances the value to 2.2 ± 0.2 , and there is little influence of an additional methoxy group in the 5'-position as in malvidin (2.1 ± 0.1). Glycosylation of anthocyanins in the 3 position diminishes the antioxidant activity to a similar extent as was shown previously with rutin (quercetin-3-rutinoside). Cyanidin-3-rutinoside (keracyanin) and the 3-O-galactoside (ideain) have TEAC values around 3 mM (Fig. 10). Malvidin glycosylation in the 3 position (oenin) re-

duces the value only slightly presumably because this structural feature does not make a significant contribution without the dihydroxy structure in the B ring. Trihydroxylation of the B ring of cyanidin as in delphinidin (Fig. 10) neither enhances nor diminishes the TEAC. This is in contrast with the situation with myricetin (with the same hydroxyl arrangement as delphinidin) vs. quercetin with the unsaturated C ring, the 4-oxo feature and the 3'-hydroxyl group or with epigallocatechin (with the same hydroxyl arrangement as myricetin and delphinidin but with a saturated heterocyclic ring).

As mentioned previously, there is great interest in the question as to whether the polyphenolic constituents of red wine are an important factor contributing to protection from coronary heart disease. Grapes and wine contain large amounts of polyphenols at high concentrations in the range of 1.0–1.8 g/ml.^{80–82} The antioxidant activity of red wine containing the blue grapeskin flavonoids has been shown in protection of LDL against oxidation and with a greater efficacy than α -tocopherol.^{83–85}

The total antioxidant activities of a range of red wines against radicals in the aqueous phase are shown in Fig. 11, with the ratio of the total antioxidant activity to the total phenol level. While the total antioxidant activities vary over a factor of 2, from 12 to 24, for the red wines, the ratios indicate the direct relationship of the antioxidant activity to the phenolic content. The comparative responses of Bouzy Rouge and the Champagne (NV) were studied because they were produced from the same grape. The differential levels of the phenolic constituents of red and white wines as mean values from fourteen reds and six whites measured by Frankel *et al.*⁸⁶ are shown in Table 6. The mean total antioxidant activity has been determined from the calculated antioxidant activities of the individual constituents and, on the basis of the composition of the individual constituents, the contribution to the total antioxidant activity is calculated (Table 7). Based on Frankel's reported figures, the calculated antioxidant activity is only 25% of the measured value. However, Frankel's figures are low compared to the data of others,^{80–82} and the remainder of the antioxidant activity is presumably unidentified polyphenols and phenolic acids as well as polymers formed from them.

DETERMINANTS OF RADICAL SCAVENGING POTENTIAL

The interaction of polyphenols with azide radicals has been applied to investigate the relative importance of the polyphenolic hydroxyl groups;⁸⁷ azide radicals will attack phenolic compounds rather indiscriminately owing to their strong electrophilicity. Substances with

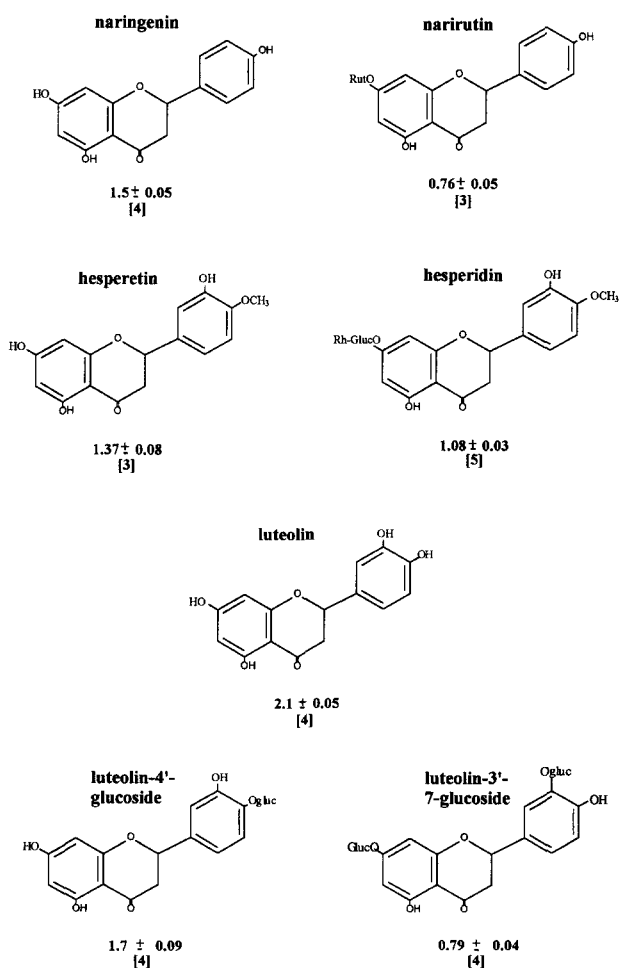


Fig. 7. Influence of glycosylation on the antioxidant activity.

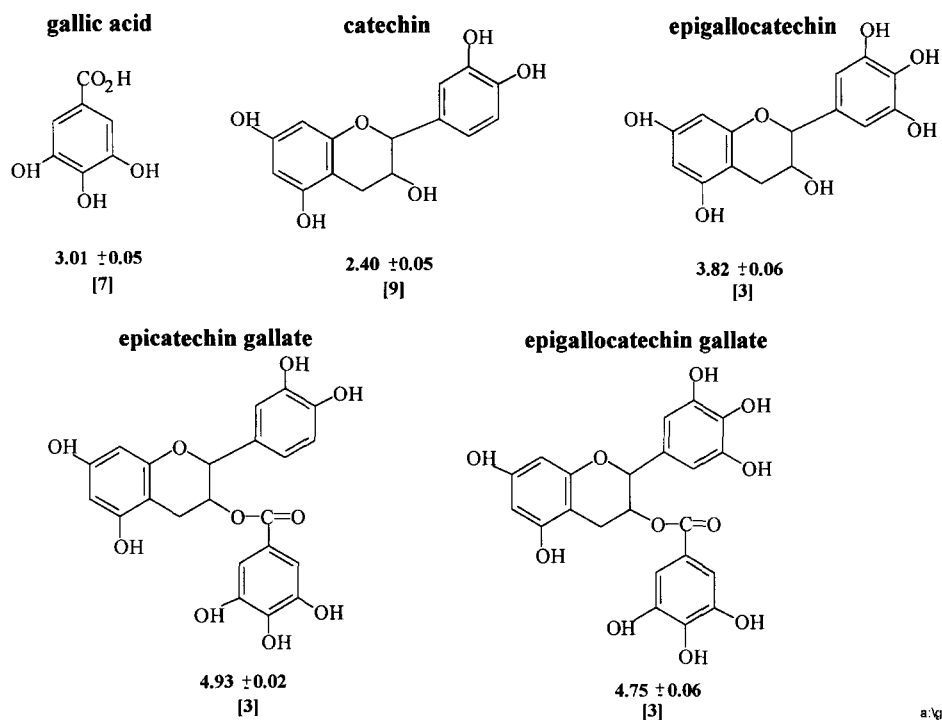


Fig. 8. The antioxidant activity of the catechins and catechin-gallate esters.

a saturated heterocyclic ring are predominantly attacked at the *o*-dihydroxy site in the B ring and the semiquinones formed are quite stable, for example, from catechin, dihydrofisetin, taxifolin (dihydroquercetin), hesperetin, cyanidin chloride. Whereas substances with a 2,3-double bond and both 3- and 5-OH substituents show extensive resonance, which does not necessarily translate into higher stability of radicals.⁸⁶

More recent studies from the same group⁵¹ have investigated the extent to which polyphenols react as radical scavengers. Applying pulse radiolysis, the rate constants with $\cdot\text{OH}$, $\text{N}_3\cdot$, $\text{O}_2^{\cdot-}$, $\text{LOO}\cdot$, $\text{tBuO}\cdot$, and sulphite have been determined as well as the stability of the antioxidant radical. The conclusion drawn was that the three criteria for effective radical scavenging^{51,88} are:

1. the *o*-dihydroxy structure in the B ring,⁸⁹ which confers higher stability to the radical form and participates in electron delocalization;
2. the 2,3 double bond in conjugation with a 4-oxo function in the C ring is responsible for electron delocalization from the B ring—the antioxidant potency is related to structure in terms of electron delocalization of the aromatic nucleus. Where these compounds react with free radicals, the phenoxyl radicals produced are stabilized by the resonance effect of the aromatic nucleus;
3. the 3- and 5-OH groups with 4-oxo function in

A and C rings are required for maximum radical scavenging potential.

Thus, quercetin, for example, satisfies all the above-mentioned determinants and is a more effective antioxidant than the flavanols (e.g., catechin), which lack aspects of the structural advantages of quercetin and other flavonols and only satisfy determinant [1]. Our findings, detailed in the Antioxidant Potentials of Polyphenols Against Radicals Generated in the Aqueous Phase and Structure–Activity Relationships section, applying the assessment of the relative ABTS^{•+} radical scavenging abilities of the flavonoid families and the interpretations of the relative values are entirely consistent with these criteria. Other approaches^{72,90} also have established that the position and degree of hydroxylation is fundamental to the antioxidant activity of flavonoids, particularly in terms of the *o*-dihydroxylation of the B ring, the carbonyl at position 4, and a free hydroxyl group at positions 3 and/or 5 in the C and A rings, respectively. It has been suggested that *o*-dihydroxy grouping on one ring and the *p*-dihydroxy grouping on the other (e.g., 3,5,8,3',4' and 3,7,8,2',5'-pentahydroxyflavones) produce very potent antioxidants, while 5,7 hydroxylation in the A ring has little influence. However, our findings on the total antioxidant activity in the aqueous phase, as shown previously, suggest that the latter is important to the antioxidant potential, but this might be less so in the lipophilic phase.

Table 4. Contribution of the Polyphenolic Flavanol Constituents to the Antioxidant Activity of Green Tea

Polyphenol	EGCG	ECG	EGC	EC	C		
%Composition (dry weight) of green tea extract	11.16	2.25	10.32	2.45	0.53	= 26.71%	(1)
Antioxidant activity (mM) (measured) (from Fig. 8)	4.8	4.9	3.8	2.5	2.4		(2)
Concentration equivalent in ppm to 1 mM	458	442	306	290	290		(3)
Contribution to antioxidant activity (calculated)	1.17	0.25	1.28	0.21	0.04	= 2.95	(4)
Actual % contribution to antioxidant activity of green tea.	32%	7%	34%	6%	1%		(5)

The contribution of each constituent to the antioxidant activity was calculated as follows: line (1) depicts the catechin/gallate composition of the green tea extract as % dry weight (from Table 2); line (2) shows the measured antioxidant activity of the individual catechin/gallate constituents for 1 mM concentration of each (as in Table 1); line (3) is the equivalent concentration of 1 mM for each constituent in ppm; line (4) is the calculated contribution of each constituent to the measured antioxidant activity of green tea (3.78 for 1000 ppm) derived from the relative proportions in line (1); line (5) is the calculated percentage contribution of each constituent to the measured antioxidant activity of green tea.

Structural features over and above the polyhydroxylic substitution of the compounds also underlie their biological activities other than antioxidant properties. The inhibitory effects on rat liver glutathione-S-transferase isoforms were dependent on the absence of a sugar moiety, on the attachment of the B ring to the 2 position of the C ring (i.e., not the isoflavones) and on unsaturation in the heterocyclic C ring at the 2-3 positions.⁹¹

Earlier studies indicated that the ability of flavonoids to inactivate peroxy radicals was in the main better than the small phenolic antioxidants, butylated hydroxyanisole, and butylated hydroxytoluene.⁹² A 2-electron oxidation was postulated with 3',4'-dihydroxy flavonols reducing peroxy radicals to produce quinones via the flavonoid phenoxyl radical). Overall, the reduction potentials of flavonoid radicals are lower than those of alkylperoxy and superoxide radicals; thus, flavonoids may inactivate these damaging oxyl species and prevent the deleterious consequences of their reaction. Furthermore, the reduction potential of Trolox is lower than those of the flavonoid radicals, which means that oxidation of vitamin E by flavonoid radicals is thermodynamically feasible.

On the basis of the rather low reduction potentials⁹³ of the flavonoid phenoxyl radicals (similar to or lower than Trolox C radical at pH 13.5), it was assumed that they are as least as effective as Trolox as hydrogen donors. Flavonoid phenoxyl radicals have been generated by bromide radical ion-induced oxidation of flavonoids

in aqueous solution to investigate the structure-reactivity relationships.⁵⁴ These workers deduced that the reduction potential of the phenoxyl radicals of catechin and rutin, for example, are lower than for hesperidin because of the electron-donating 3'-O-substituent, that for catechin being lower than rutin because of the absence of the —CH=CH— bond.

Bors *et al.*⁹⁴ have suggested that the stability of flavonoid aryloxyl radical is sometimes questionable and may give rise to pro-oxidant effects. This might help explain the occasional, unpredictable relationships sometimes observed between the structure of some flavonoids and their antioxidant activities. Kinetic modeling has been applied to measure the relative rate constants for the reaction of a range of flavonoids with azide radicals generated by pulse radiolysis. These investigators propose that the two structural features that control the redox potentials are the catechol group in the B ring and the 2-3 double bond in the C ring. Thus, all substances containing the above structural features were found to have a higher redox potential than ascorbate and were capable of oxidising it to the ascorbyl radical, and quercetin belongs to this group. However, taxifolin has a lower redox potential than the ascorbyl radical and it might be expected that hesperidin and naringenin belong to this group. This might be important in considering the protection of aryloxyl radicals from degradation as well as in terms of the synergistic interactions of these antioxidants. These chemical considerations are exemplified in biological observations of the enhancement of the antiproliferative effect of quercetin and fisetin by ascorbic acid due to its ability to protect the polyphenols against oxidative degradation.⁹⁵

Table 5. Total Antioxidant Activity of Green Tea

	TAA mM
Green tea (1000 ppm)	3.78 ± 0.03
Green tea polyphenol extract (44% composition)	3.36 ± 0.07
Combined pure catechin constituents proportionately combined	2.76 ± 0.06
Catechin substituents (calculated) ^a	2.95

^a Based on TEAC values for individual catechins.

ANTIOXIDANT ACTIVITY AGAINST RADICALS GENERATED IN THE LIPOPHILIC PHASE

Many studies have been undertaken in lipophilic systems to establish the structural criteria for the activity of polyhydroxy flavonoids in enhancing the stability

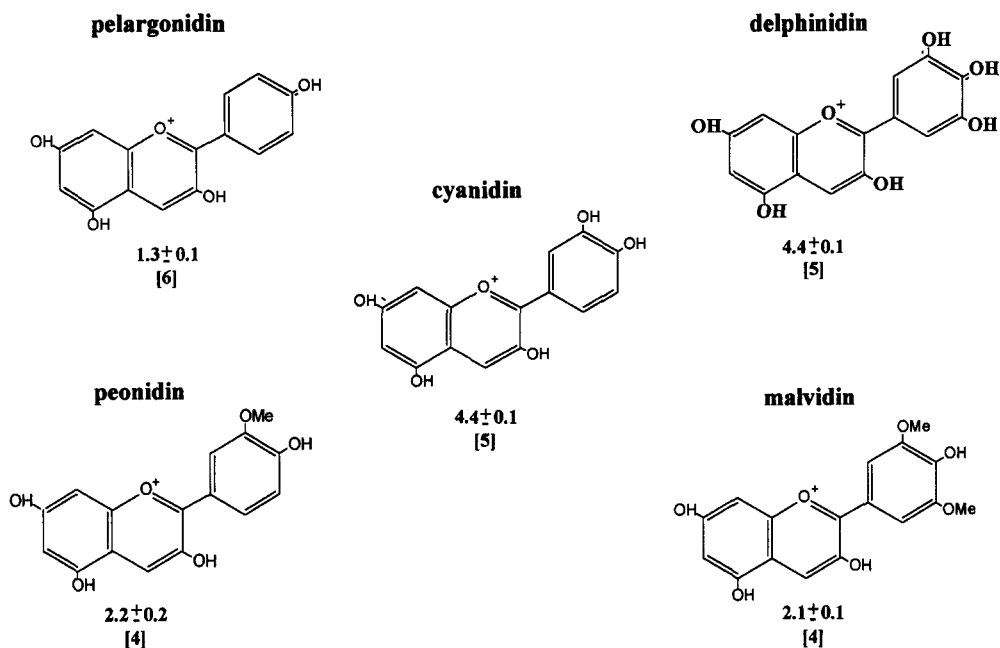
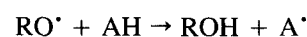
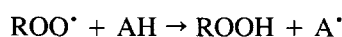


Fig. 9. The effect of variations in the B ring on the antioxidant activities of the anthocyanidins.

of fatty acid dispersions (especially methyl linoleate), lipids, oils, low density lipoproteins, and lard towards oxidation^{12,49,57,59,71,73,79,90,96} The specific mode of inhibition of oxidation by the individual polyphenols is not clear but they may act by: a) chelating copper ions via the ortho dihydroxy phenolic structure; b) scavenging lipid alkoxyl and peroxy radicals by acting as chain breaking antioxidants, as hydrogen donors



and c) regenerating α -tocopherol through reduction of the α -tocoperoxyl radical.

The phenoxyl radical formed by reaction of a phe-

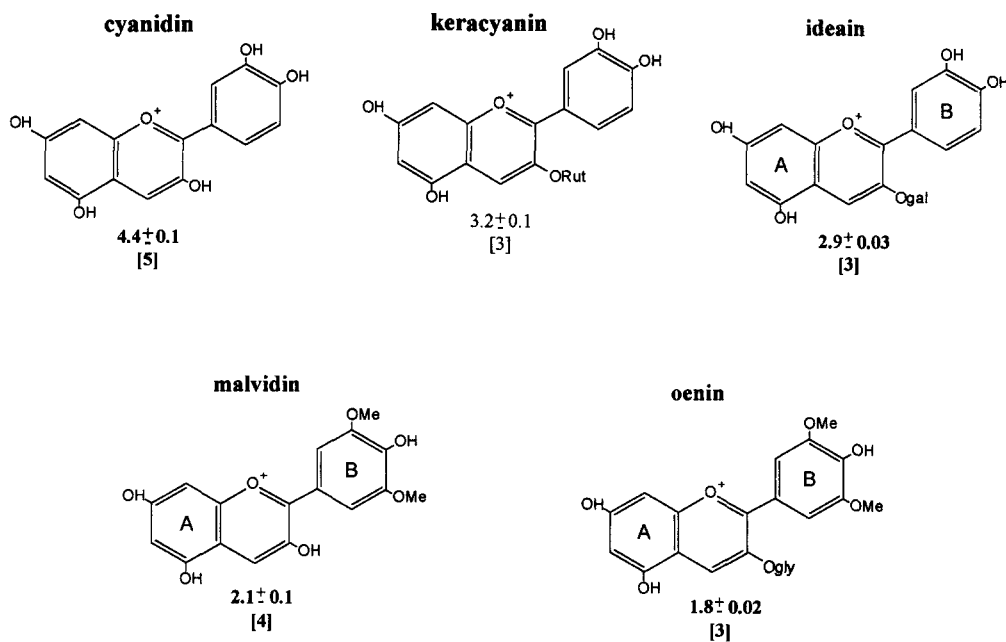


Fig. 10. Antioxidant activity of anthocyanidins and anthocyanins.

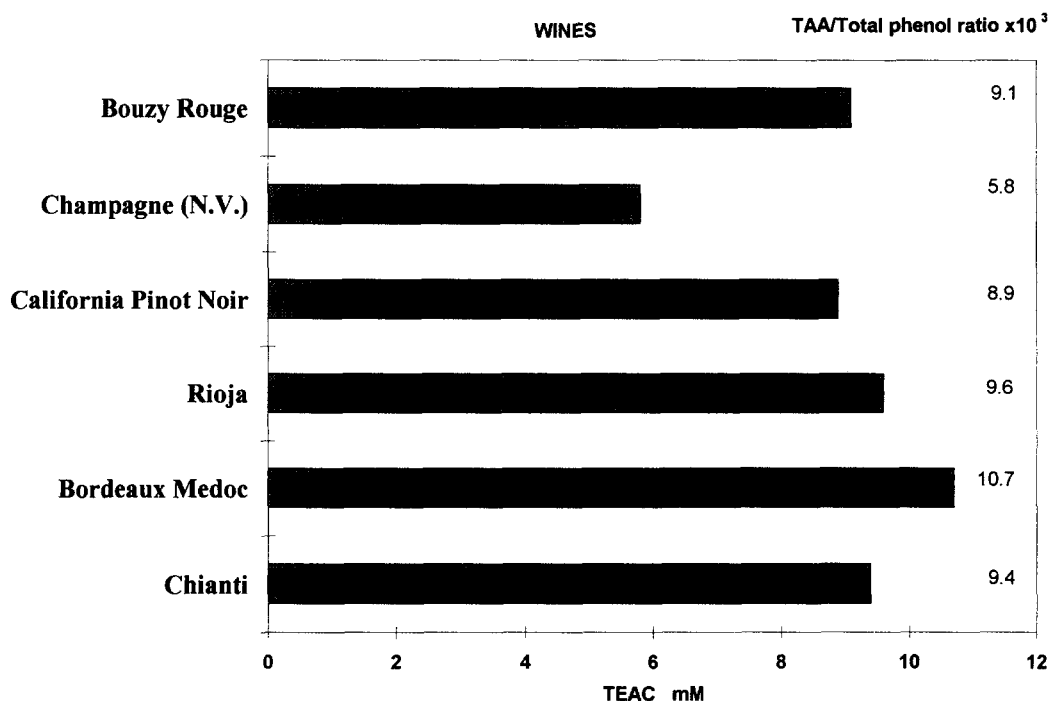


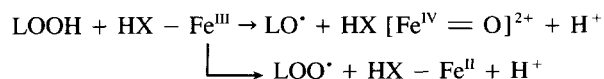
Fig. 11. Total antioxidant activity of red wines and their total phenol content. Bouzy rouge—1986; Champagne blanc de noir (from the same grape as bouzy rouge); Pinot Noir—California, 1991; Rioja—1990; Bordeaux Medoc—1989; Chianti classico—1990.

nolic antioxidant with a lipid radical is stabilized by delocalization of unpaired electrons around the aromatic ring. The *o*-dihydroxy substitution in the B ring is important for stabilizing the resulting free radical form. The possibility exists for stabilization of radical forms through the 3-OH, 5-OH, and 4-oxo groups and conjugation from the A ring to the B ring through the additional, 2,3 unsaturation in the C ring. This type of reaction is mainly seen with aliphatic peroxy radicals reacting with phenolic antioxidants, as pointed out by Bors *et al.*⁵¹

Comparison of a range of flavanones and flavones in their capacity to increase the induction period to autoxidation of fats has led to the conclusion that optimum antioxidant activity is associated with such structural features⁹⁶ as: multiple phenolic groups, especially the 3',4'-orthodihydroxy configuration in the B ring; the 4-carbonyl group in the C ring. However, in contrast with aqueous phase interactions, the 2,3-double bond is deemed less important because taxifolin is more effective than its unsaturated analog quercetin. Catechin, lacking the 4-carbonyl group as well as the 2,3 double bond, is also relatively ineffective. A free 3-OH group or 3- and 5-OH groups present simultaneously are also considered to be important in the lipophilic phase; thus, luteolin, which lacks the 3-OH but relies on the 5-OH in the A ring with the 4-carbonyl groups in the C ring, was found to be less effective

than quercetin and related flavonols. However, it is not clear whether these differential structural features and their influence as antioxidants can be totally ascribed to a hydrogen-donating antioxidant effect or whether the partition coefficients of the compounds into the lipophilic region and their accessibility to the autoxidising lipids has confounded these effects.

The oxidation of low density lipoproteins can be used as a model for investigating the efficacy of the polyphenols as chain-breaking antioxidants. Free radical-mediated peroxidation of polyunsaturated fatty acids leads to the formation of lipid hydroperoxides through a chain reaction of peroxidation. Oxidative and reductive decomposition of peroxides mediated by heme-proteins or transition metal ions can amplify the peroxidation process.



The presence of chain-breaking antioxidants can

Table 6. Phenolic Constituents of Wine^a

	Red (mg/l)	White (mg/l)
Catechin	191	35
Epicatechin	82	21
Gallic acid	95	7
Cyanidin	3	0
Malvidin-3-glucoside	24	1
Rutin	9	0
Quercetin	8	0
Myricetin	9	0
Caffeic acid	7.1	2.8
Resveratrol	1.5	0

^a From ref. 86.

intercept this peroxidation process by reducing the alkoxyl or peroxy radicals to alkoxides or hydroperoxides, respectively, the hydroperoxides reentering the cycle until the antioxidants are consumed. It has been proposed that alkoxyl radicals rearrange through their own reactivity to epoxides.⁹⁷ The oxidative interaction of LDL with heme proteins is hydroperoxide dependent,⁹⁸ and, without the addition of initiating species, these agents will slowly cycle the endogenous peroxides within the LDL and amplify the peroxidation process. To study the antioxidant activity of polyphenols as scavengers of propagating lipid peroxy radicals, no initiating species were added, but metmyoglobin was applied to propagate the decomposition of the minimal levels of endogenous preexisting lipid hydroperoxides.⁹⁸ Copper ions were avoided to eliminate the confounding effects of the polyphenols as metal-chelators.³

In LDL, on oxidation, the aldehydic decomposition products of peroxidation can be assessed as markers of the oxidation of the polyunsaturated fatty acids. They may bind to the apoprotein B100 on the surface of the LDL, specifically the amino groups, altering the charge and recognition properties, and these modifications can be monitored as changes in electrophoretic mobility as a further indication of the oxidative modification of the LDL. Thus, the extent of inhibition of LDL oxidation by the polyphenols can be assessed. The relative efficacies of the catechin/gallate polyphenols in inhibiting LDL oxidation⁷³ are in the sequence of gallic acid being the least effective, requiring about 1.2 μ M for 50% inhibition of maximal oxidation, epigallocatechin 0.75 μ M, whereas catechin, epicatechin, epicatechin gallate (ECG) and epigallocatechin gallate (EGCG), were all very similar with values ranging from 0.25 to 0.38 μ M. A similar sequence is seen in the inhibition of altered relative electrophoretic mobility, as expected. Miura et al.⁷⁹ also found that epigallocatechin was the least effective catechin in protecting LDL from copper-mediated oxidation, whereas epicatechin gallate was the most effective,

with a threefold decrease in the IC₅₀, consistent with our findings.

The catechin/gallate family of compounds was also studied for their ability to spare vitamin E and protect it from oxidation. LDL contains a number of endogenous antioxidants including α - and γ -tocopherols, β -carotene, lycopene, and other carotenoids.⁹⁹ The reduction potentials of flavonoid radicals are higher than that of Trolox, which means that their reaction with vitamin E is thermodynamically feasible.⁹³ Monitoring the consumption of vitamin E in LDL when challenged with a pro-oxidant in the form of metmyoglobin in the presence of the catechin polyphenols (2 μ M) demonstrates that epigallocatechin is, indeed, the least effective in sparing the vitamin E, reflecting its lesser contribution to increasing the resistance of LDL to oxidation, whereas the delay in consumption of the LDL-vitamin E was prolonged by epigallocatechin gallate and epicatechin gallate. Flavonoid aglycones are rather lipophilic antioxidants, although generally more hydrophilic than α -tocopherol. It has been hypothesized that catechins might be localized near the membrane surface scavenging aqueous radicals and preventing the consumption of tocopherol, whereas α -tocopherol mainly acts as a chain-breaking lipid peroxy radical scavenger within the LDL.⁵⁷

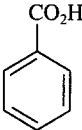
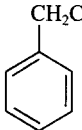
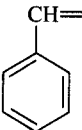
Catechin, epicatechin, and quercetin have been shown to have powerful antioxidative capacities to approximately the same extents, in phospholipid bilayers exposed to aqueous oxygen radicals,⁵⁷ although the electron-donating ability of catechin is lower than that of quercetin. On the other hand, quercetin is more effective than catechin as an antioxidant in protecting low-density lipoproteins from oxidation in copper-mediated peroxidation systems (Paganga et al., unpublished). Furthermore, these flavonoids have been shown to con-

Table 7. Contribution of Identified Constituents to the Total Antioxidant Activity of Red Wine

	TEAC (mM)	Composition ^a (mM)	Contribution to TAA
Catechin	2.4 $\pm$ 0.05	0.66	1.6
Epicatechin	2.5 $\pm$ 0.02	0.28	0.7
Gallic acid	3.01 $\pm$ 0.05	0.51	1.54
Cyanidin	4.42 $\pm$ 0.12	0.01	0.04
Malvidin-3-glucoside	1.78 $\pm$ 0.02	0.05	0.09
Rutin	2.42 $\pm$ 0.12	0.01	0.02
Quercetin	4.72 $\pm$ 0.10	0.02	0.09
Myricetin	3.72 $\pm$ 0.28	0.03	0.09
Resveratrol	2.00 $\pm$ 0.06	0.006	0.01
Caffeic acid	1.26 $\pm$ 0.01	0.014	0.02
Total contribution to TAA			4.2
Mean TEAC for red wines			16.7

^a Composition data from ref. 86.

Table 8. Total Antioxidant Activities (mM) Relative to Trolox of the Hydroxybenzoic, Hydroxyphenylacetic, and Hydroxycinnamic Acids

	Hydroxybenzoic Acids		Hydroxyphenylacetic Acids		Hydroxycinnamic Acids	
						
<i>Position of OH</i>						
2 (salicylic)	0.04 ± 0.01	[4]	0.99 ± 0.09	[5]	(<i>o</i> -coumaric)	0.99 ± 0.15 [4]
3	0.84 ± 0.05	[5]	0.90 ± 0.11	[5]	<i>m</i> -	1.21 ± 0.02 [4]
4	0.08 ± 0.01	[3]	0.34 ± 0.10	[3]	<i>p</i> -	2.22 ± 0.06 [7]
2,3	1.46 ± 0.01	[3]				
3,4 (protocatechuic)	1.19 ± 0.03	[4]	2.19 ± 0.08	[4]	(caffeic)	1.26 ± 0.01 [3]
2,5	1.04 ± 0.03	[3]	0.91 ± 0.05	[4]		
3,5 (resorcylic)	2.15 ± 0.05	[4]				
4-hydroxy, 3-methoxy	1.43 ± 0.05	[3]	1.72 ± 0.06	[3]	(ferulic)	1.90 ± 0.02 [9]
3,4,5 (gallic)	3.01 ± 0.05	[7]				
pyrogallol	1.91 ± 0.02	[6]				
gallic acid and methyl ester	2.40 ± 0.03	[3]				
3,5-dimethoxy,4-hydroxy (syringic acid)	1.36 ± 0.01	[3]				

serve endogenous α -tocopherol in LDL, and quercetin is the most effective of the compounds studied.⁵⁹ It has been proposed that flavonoids near the surface of phospholipid structures are ideally located for scavenging oxygen radicals generated in the aqueous phase. Yuting *et al.*⁴⁹ have applied a system of inhibition of lipid peroxidation in mouse liver homogenates as a marker of antioxidant efficiency, with IC_{50} (μ M) values: rutin [3.2], morin [4.4], quercetin [5.2], acetin [14], hispidulin [64], and for naringin and hesperidin no inhibition. It is difficult to envisage a relationship between structure and activity from this study either on the grounds of the structural criteria defined above or a hypothesis of potential abilities to scavenge peroxy radicals or relative solubilities in the lipid phase.

PHENOLIC ACIDS

The antioxidant activity of phenolic acids and their esters depends on the number of hydroxyl groups in the molecule that would be strengthened by steric hindrance.⁹⁰ The electron-withdrawing properties of the carboxylate group in benzoic acids has a negative influence on the H-donating abilities of the hydroxy benzoates. Hydroxylated cinnamates are more effective than benzoate counterparts.

Hydroxybenzoic acids

The monohydroxy benzoic acids show no antioxidant activity in the *ortho*, and *para* positions in terms of hydrogen-donating capacity against radicals generated in the aqueous phase but the *m*-hydroxy acid has an antioxidant activity of 0.84 ± 0.05 mM (Table 8). This is consistent with the electron withdrawing

potential of the single carboxyl functional group on the phenol ring affecting the *o*- and *p*-positions. The monohydroxybenzoates are, however, effective hydroxyl radical scavengers,¹⁰⁰ due to their propensity to hydroxylation and the high reactivity of the hydroxyl radical. With a methylene group between the phenolic ring and the carboxylate group, as in the phenylacetic acids, the *o*- and *m*-hydroxy derivatives have antioxidant activities close to 1 mM, while the activity of *p*-hydroxy phenylacetic acid is only slightly enhanced.

The dihydroxybenzoic acid derivatives show an antioxidant response dependent on the relative positions of the hydroxyl groups in the ring. Dihydroxylation in the *ortho* and *meta* positions to the carboxylate group, 2,3-dihydroxy benzoic acid, gives a TEAC value of 1.46 mM relative to Trolox or α -tocopherol at 1.0, which is slightly elevated compared to the *meta*, *para* disubstitution in 3,4-dihydroxy benzoic acid (protocatechuic acid) with a TEAC value of 1.2 mM. With both hydroxyl substituents *ortho* to the carboxylate group in 2,5-dihydroxy benzoic acid the value is approximately the same at 1.1 mM. Thus, the proximity of the $-\text{CO}_2\text{H}$ to the orthodiphenolic substituents apparently influences the availability of the hydrogens with the *m*-position being the most effective. 3,5-Dihydroxybenzoic acid (resorcylic acid) (TEAC 2.15 mM) shows a very much enhanced Trolox equivalent antioxidant activity, which is comparable to the value for resorcinol, the equivalent compound without the CO_2H , showing the lesser influence of the electron withdrawing potential of this substituent when not adjacent to the hydroxyl groups. Incorporation of an additional $-\text{CH}_2-$ between the phenyl ring and the car-

boxylic acid group in the hydroxyphenyl acetic acids decreases the impact of the carboxylate group and almost doubles the antioxidant capacity (2.2 mM) of homoprotocatechuic acid compared with the benzoic acid derivative.

Gallic acid, the 3,4,5-trihydroxy benzoic acid, has an antioxidant capacity of 3.0 mM, corresponding to the three available hydroxyl groups. Esterification of the carboxylate group of gallic acid also decreases the effectiveness (2.4 mM). While substitution of the 3- and 5-hydroxyl with methoxy groups in syringic acid demonstrates a diminution in antioxidant activity (1.36 mM) compared to the trihydroxy derivative, the presence of the two methoxy groups adjacent to the OH group in *p*-hydroxybenzoic acid grossly enhances the hydrogen availability. It is interesting to note that insertion of an additional hydroxyl group into resorcinol (1,3-benzenediol) in the 2 position to produce pyrogallol decreases the overall antioxidant capacity (1.91 mM), showing the dampening effect of the adjacent trihydroxy sequence on the antioxidant abilities of the dihydroxy structure when the hydroxyl groups are meta. Interestingly, in contrast with flavonoids such as luteolin, morin and catechin, an antioxidant effect of gallic acid on carbon tetrachloride-induced microsomal lipid peroxidation was not detected.¹⁰¹ This raises the interesting issue when assessing antioxidant activities against lipid oxidation of the relative contributions from direct scavenging of the initiating species, the rate constant for peroxy radical scavenging and the partitioning effects into the membrane.

Hydroxycinnamic acids

Among the most widely distributed phenylpropanoids in plant tissues are the hydroxycinnamic acids, coumaric, caffeic and ferulic produced from the shikimate pathway from L-phenylalanine or L-tyrosine (Fig. 12).

Insertion of an ethylenic group between a phenyl ring carrying a *p*-hydroxyl group and the carboxylate group, as in *p*-coumaric acid, has a highly favourable effect on the reducing properties of the OH group (TEAC 2.2) compared with cinnamic acid (0) (and *p*-hydroxyphenyl acetic acid 0.34), whereas the equivalent *m*- and *o*-coumaric acids have TEAC values closer to unity (Table 8). Incorporation of a hydroxyl group into *p*-coumaric acid adjacent to that in the *para* position as in caffeic acid gives a TEAC of 1.26 mM. Comparing the cinnamates with the phenylacetic acid derivatives, this value is considerably lower than that of 3,4-dihydroxyphenylacetic acid (2.19 mM). This is consistent with the electron donating effects on the ring of the COOH—CH=CH— vs. COOH—CH₂— groups and the relationship with the number and posi-

tion of hydroxyl groups in the ring, the monohydroxyl group in the cinnamic acids being more available as hydrogen donors than the monohydroxyl groups in the phenylacetic acid. On the other hand, dihydroxylation in the 3,4 position enhances the efficacy of the latter while decreasing that of the *p*-coumaric acid. In fact, the antioxidant activity of caffeic acid (3,4-dihydroxycinnamic acid) (1.26 mM) is almost the same as that of protocatechuic acid (3,4-dihydroxybenzoic acid). Glycosylation of the carboxylate group of caffeic acid (chlorogenic acid) has no influence on the TEAC value, 1.24 (as expected), and their spectra have a closer structural features in terms of band resolution than with ferulic acid. Substitution of the 3-hydroxyl group of caffeic acid by a methoxy group (ferulic acid) considerably enhances the antioxidant effectiveness of this (1.9 mM). This can be compared with the 4-hydroxy, 3-methoxy derivative of phenylacetic acid, homovanillic acid, with a similar value of 1.72 mM, in contrast with the lower value for the benzoic acid derivative, vanillic acid, (1.43 mM) influenced by the adjacency of the carboxylate groups to the phenyl ring.

Investigation of the antioxidant potential of phenolic acids in lipophilic systems consisting of accelerated autoxidation of methyl linoleate under conditions of intensive oxygenation at 110°C for several hours has been undertaken.¹⁰² All the monophenols apart from BHA are less effective than polyphenols. Introduction of a second hydroxyl group in the *ortho* position—caffeic—or *para* position—protocatechuic acid—enhances the antioxidant activity^{71,103} in lipid systems, making these phenolic acids more efficient than their respective monophenols *p*-hydroxy benzoic acid and *p*-coumaric acid. The results from these lipid studies also show that the antioxidative efficiency of monophenols is increased substantially by one or two methoxy substitutions in positions *ortho* to the OH as in ferulic acid: sinapic acid is more protective than ferulic acid, which is better than *p*-coumaric acid, and syringic acid is more active than vanillic and *p*-hydroxybenzoic acids. At least two, or even three, neighboring phenolic hydroxyl groups and a carbonyl group in the form of an aromatic ester, *o*-lactone, or a chalcone, flavanone or flavone are essential molecular features required to achieve a high level of antioxidant activity.⁹⁰ However, in the aqueous phase ferulic acid is 150% as efficient as caffeic and chlorogenic acids.¹⁰⁴

Several investigations^{71,103,105} have shown that *ortho* substitution with electron-donating alkyl or methoxy groups increases the stability of the aryloxy radical and so its antioxidant potential. Indeed, the methyl groups *para* to the functional hydroxyl groups in the chromanol ring of vitamin E are fundamental to the hydrogen donating ability (the same applies to butylated hydroxytoluene).

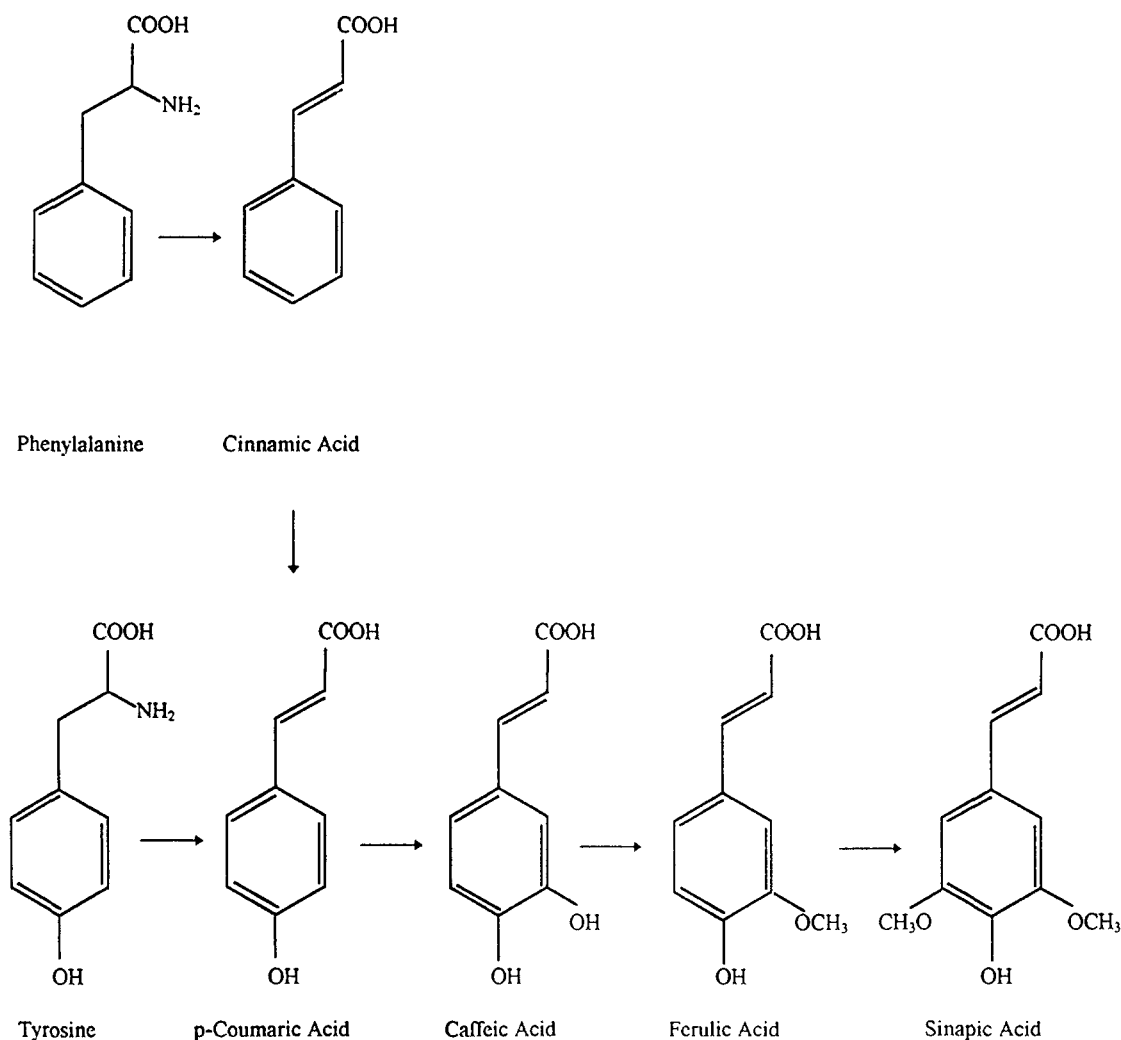


Fig. 12. Biosynthetic pathway of the phenylpropanoids in plants.

The relative antioxidant activities of the hydroxycinnamates against propagating peroxy radicals generated in the lipophilic phase of LDL as described in the Antioxidant Activity Against Radicals Generated in the Lipophilic Phase section have been examined.⁶⁵ The concentration for 50% inhibition of LDL oxidation is less for chlorogenic acid and caffeic acid ($0.33 \mu\text{M}$) than ferulic acid ($0.9 \mu\text{M}$) and *p*-coumaric acid ($4 \mu\text{M}$). A similar sequence is noted for resistance of LDL cholesterol to oxidation. Thus, the diphenolics, chlorogenic and caffeic acids, apparently have a higher radical scavenging ability than monophenolics (*p*-coumaric acid) consistent with the chemical criteria applied to diphenolics.⁵¹ Methoxylation of the hydroxyl group in the *ortho* position of the diphenolics, as in ferulic acid, results in a decrease in the scavenging reaction, for instance, hydroxylation as in caffeic acid in place of methoxylation is substantially more effective. Ferulic acid is, indeed, expected to be more effective

than *p*-coumaric acid because the electron-donating methoxy group allows increased stabilization of the resulting aryloxy radical through electron delocalization after hydrogen donation by the hydroxyl group.

The order of scavenging effectiveness observed here in LDL is consistent with previous observations,¹⁰² proposing that the antioxidant efficiency of monohydroxycinnamates, such as *p*-coumaric acid, against autoxidation of methyl linoleate under strongly oxidising conditions is increased substantially by methoxy substitution in positions *ortho* to the hydroxyl group, as in ferulic acid. Other studies on the effects of hydroxycinnamates on the induction period of autoxidizing fats have also demonstrated the order of effectiveness caffeic > ferulic > *p*-coumaric acid.¹² Phenolic substances are the most effective antioxidants from natural sources including alkoxy phenols that contain one free and one alkylated hydroxyl group, usual methoxy, or polyphenols with *ortho*- or *para*-dihydroxylic

groups, or phenols containing condensed rings, for example, anthocyanins, flavones. Ferulic, caffeic, and chlorogenic acids fall into the former two categories. The mechanism of action of phenolic acids as antioxidants has been proposed from studies on the autoxidation of linoleic acid micelles¹⁰³ through the direct inhibition of *trans,trans*-conjugated diene hydroperoxide formation related to the H-donating ability of the phenol.¹⁰⁵ In addition, ferulic acid (reviewed in¹⁰⁶) has been shown to retard the peroxidation of linoleic acid.¹⁰⁷ In the case of the hydroxybenzoic acids, however, methoxy substitution is far from equivalent to the addition of an OH group—ferulic acid and vanillic acid remained substantially less efficient than caffeic acid and protocatechuic, respectively, in the lipophilic phase for methyl linoleate¹⁰² and for LDL oxidation,⁶⁵ but the accessibility of the antioxidant to the peroxy radical, or its partition coefficient, is an additional consideration here that might influence the efficacy.

The presence of the $-\text{CH}=\text{CH}-\text{COOH}$ groups in cinnamic acid ensures greater H-donating ability and subsequent radical stabilization than the carboxylate group in benzoic acids. The reduction potentials of radicals derived from 3,4-dihydroxybenzoate derivating decrease with the electron-donating power at C1, for example, dihydroxybenzoate radicals, have a higher reduction potential than dihydroxycinnamate radicals.⁵⁴ Thus, caffeic, sinapic, ferulic, and *p*-coumaric acids were found to be more active than protocatechuic, syringic, vanillic, and *p*-hydroxybenzoate, especially in lipid systems.¹⁰² It may be that the $-\text{C}=\text{C}-\text{CO}_2\text{H}$ linked to the phenyl ring plays a role in stabilizing the radical by resonance and, hence, caffeic acid is more efficient than pyrocatechol. On the other hand, the fact that protocatechuic acid was found to be less efficient than pyrocatechol tends to demonstrate a negative impact of a carboxylate group direct linked to the phenyl ring, in this instance, in contrast to observations of the relative scavenging activities of resorcinol and resorcylic acid, and pyrogallol and gallic acid in the aqueous phase, as mentioned above. Cuvelier et al.¹⁰² suggested a decreasing effect of esterification in the lipid phase, and this was our observation in the aqueous phase for gallic acid methyl esters, but our studies revealed no differences between caffeic acid and its quinic acid ester, chlorogenic acid, in their inhibitory effects on LDL oxidation; it may be that this can be ascribed to increased hydrophilicity of the compounds.

CONCLUSIONS

It is possible, based on the foregoing, to assemble a hierarchy of antioxidant potential against radicals generated in the aqueous phase relative to that of Tro-

lox. Tables 3 and 8 summarizes the findings presented here for the flavonoids. The values of antioxidant activity determined by TEAC method reflect the range of structures of the compounds and are consistent with the criteria of Bors et al.,⁵¹ Dziedic and Hudson,⁹⁰ and the experimental findings of others in aqueous systems. There is also broad agreement between our findings in oxidising LDL systems and those of others in a range of lipid systems or in the presence of transition metal inducers of lipid oxidation, although it must be emphasized that many different systems have been applied, which makes appropriate comparisons difficult overall. This might reflect the influence of the partition coefficients of the compounds on the accessibility to radicals in the lipophilic phase, per se, rather than the direct consideration of the rates of the scavenging reactions. Indeed, a recent report¹⁰⁸ has studied the order of efficacy of quercetin, rutin, hesperetin, and naringenin as antioxidants in three different types of preparation: a linoleic acid peroxidation system compared with a rat cerebral membrane autoxidation system and with dipalmitoyl phosphatidyl choline vesicles. The authors ascribe the conflicting sequences of antioxidant potential in the different systems to their differential abilities to penetrate and interact with the lipid bilayers.

Little is known about the bioavailability, absorption and metabolism of the polyphenols in humans, and it is likely that different groups of flavonoids have different pharmacokinetic properties. Evidence exists that catechin is absorbed by the human gut¹⁰⁹ and for the absorption of catechin in male volunteers following oral administration of radiolabelled 3-O-methyl catechin.¹¹⁰ The major urinary metabolites were glucuronides of 3-3-O-dimethyl catechin and a glucuronide and a sulphate of 3-O-methyl-catechin. Plasma levels of the latter reached a peak within 2 h of administration. It has been proposed that quercetin might reach levels of up to 1 μM in human plasma,¹¹¹ although the findings are conflicting. Others studies have reported that quercetin after oral administration in humans is not detected in plasma or urine, but $\approx 50\%$ is recovered in the feces, suggestive of extensive degradation.¹¹² Dietary studies in rats have shown a reduction in mammary tumors¹¹³ and suggested only 20% is absorbed from the GI tract.¹¹⁴ Others report that the major portion of ingested flavonoids (44%) is present in the gastrointestinal tract before excretion in the bile. It was suggested in 1959 that flavonoids are rapidly absorbed and converted to a variety of hydroxyaromatic acids, which are rapidly eliminated in the urine.¹¹⁵ The metabolism of quercetin (Fig. 13), for example, is proposed to be via the splitting of the flavonoid molecule to form hydroxyaromatic acids with a two-carbon side chain because a hydroxyl group is on the 3-carbon of the pyrone ring; when a hydroxyl group is lacking in this position, as

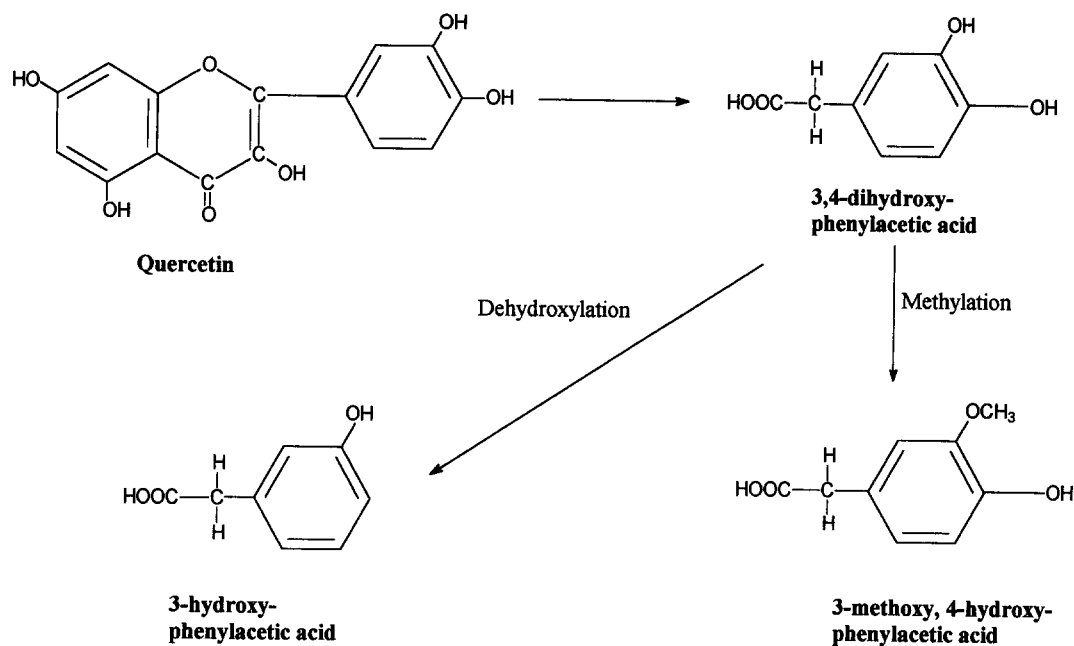


Fig. 13. Structural representation of quercetin metabolism.

in hesperidin, hydroxyaromatic acids with three-carbon side chains are formed. These latter compounds undergo β -oxidation to benzoic acid derivatives. The metabolic transformation of caffeic acid (Fig. 14) also involves the processes of methylation of the phenolic hydroxyl group, dehydroxylation in the *para* position, hydrogenation of the side chain, β -oxidation and formation of conjugates, as in the metabolism of the flavonoids. However, there is a paucity of information on the handling of these compounds by the gastrointestinal tract, their absorption, metabolism, and excretion in humans, and much work further work needs to be done in this area. It is also of interest to note that much early evidence suggests absorption of the phenolics with one aromatic ring and raises the issue of scission of the larger flavonoids in the gastrointestinal tract to lower molecular weight forms prior to absorption.

Another major factor influencing uptake may well be the interaction of polyphenols with other molecular species such as proteins, arising from the fact that polyphenols are multidentate ligands able to bind simultaneously at more than one point to the protein surface.¹¹⁶ In particular, complementarity between the polyphenol multidentate ligand and the protein is maximized by conformational flexibility in both components,¹¹⁷ proline-rich residues having a higher affinity for polyphenols. Water-soluble polyphenols (catechins, etc.) have a substantially weaker affinity for proteins than those that are more poorly water soluble (e.g., quercetin). Furthermore, in this context, note should be taken of the fact that the orthodihydroxy phenolic catechol-type

structure in the B ring has a high affinity for iron chelation. Thus, the question of the interference with iron absorption by the polyphenol or vice versa must be addressed in addition to the issue of free radical-mediated damage to gut tissue in the simultaneous presence of iron and polyphenol.

It should also be emphasized, however, that certain flavonoids in chemical systems autoxidize readily, especially quercetin, myricetin, quercetagenin, and delphinidin.¹¹⁸ More recent studies from the same group have shown the inhibition of mitochondrial respiration and cyanide-stimulated generation of reactive oxygen species by certain flavonoids.¹¹⁹ In particular, inhibition of complex I (NADH-coenzyme Q reductase) was effected in the sequence robinetin > rhamnetin >>>> 7,8-dihydroxy flavone was observed. The conclusions drawn are that flavonoids with adjacent tri-hydroxyl or *para*-dihydroxyl groups exhibit a substantial rate of autoxidation generating superoxide and hydrogen peroxide, which is accelerated by cyanide addition; flavonoids possessing the dihydroxy catechol structure exhibited a slow rate of autoxidation, also further stimulated by cyanide. Furthermore, there are reports of the prooxidant activity of some polyphenols in the presence of metal ions in which high concentrations (25–100 μ M) accelerate hydroxyl radical formation and DNA damage in vitro, mediated by iron-EDTA but not by iron-ADP or iron (II) added as ferrous sulphate.^{120,121} This is surprising, in view of the fact that another property of polyphenols, depending on their precise structure and the proximity or adja-

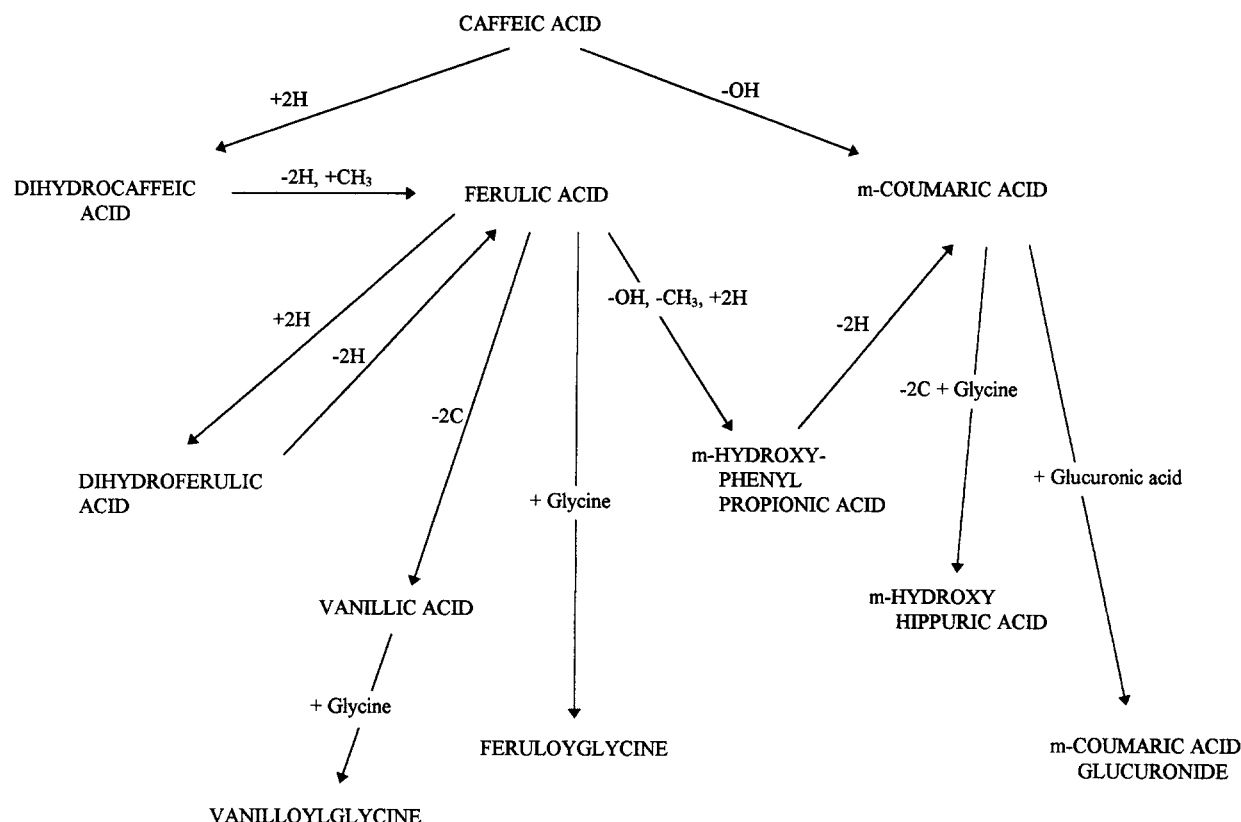


Fig. 14. Structural representation of caffeic acid metabolism.

cence of hydroxyl groups, is their metal-chelating potential.^{120,122} Thus, polyphenols may also have the possibility of chelating metal ions and preventing iron- and copper-catalyzed formation of initiating radical species. It has been proposed that the two possible points of attachment of transition metal ions to the flavonoid molecule are the *o*-diphenolic groups in the 3',4'-dihydroxy position in ring B and the ketol structure, 4-oxo, 3-OH or 4-oxo, 5-OH in the C ring of the flavonols, although it is likely that different metals show quite different properties with regard to chelation by flavonoids. High concentrations (100 μ M) of flavonols, myricetin and gossypetin, can also modify LDL through processes involving the covalent modification of the apoB₁₀₀ protein;¹²³ these studies involved very high relative concentrations of flavonoid:LDL. It is unlikely that these compounds achieve such high concentrations in vivo.

However, the antioxidant actions of myricetin and quercetin are proposed to be responsible for part of the beneficial effects of *Ginkgo biloba* on brain neurons subject to ischemia though a reduction in oxidative metabolism without affecting intracellular calcium concentrations.¹²⁴ A study of the genotoxic potential of the flavonoids quercetin, apigenin, luteolin, rhamnetin, and isorhamnetin, have been investigated

using bacterial assays. No DNA damage was detected in the "repair test" in *Salmonella typhimurium*, while the flavonoids only weakly induced the SOS test system in *E. coli* K-12 strains. The addition of a liver activation system did not increase the mutagenic effect of the flavonoids tested.¹²⁵ Dietary polyphenols have also been shown to protect mammalian and bacterial cells from cytotoxicity induced by hydrogen peroxide, especially compounds with the orthodihydroxy phenolic structure quercetin, catechin, gallic acid ester, caffeic acid ester.^{126,127} However, in contrast, ferulic acid and α -tocopherol were ineffective.

The studies reported here along with those from other laboratories may help to identify the active ingredients in beverages, vegetables, grains, and fruit (once their absorption characteristics are identified) that may protect against free radical damage, LDL oxidation, implicated in the pathogenesis of coronary heart disease, platelet aggregation, and endothelium-dependent vasodilatation of the arteries, as well as against DNA damage and cancer. This might be useful information from the point of view of identifying appropriate foods that are rich in these protective components for the development of safe food products and additives with appropriate antioxidant properties.

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